

Minimization, Mitigation, and Monitoring Plan

Port Everglades Sand Bypassing Project Broward County, FL

FDEP Permit No. 0289308-004-JC

USACE Permit No. SAJ-2008-02034 (SP-LCK)

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1.0 INTRODUCTION

This document provides minimization and mitigation requirements and biological monitoring protocols for the Port Everglades Sand Bypass Project, authorized by Florida Department of Environmental Protection (FDEP) **Permit No. 0289308-004-JC** and United States Army Corps of Engineers (USACE) **Permit No. SAJ-2008-02034 (SP-LCK)**. The project consists of the construction of an approximately 10.8-acre sediment impoundment basin on the north side of the outer Port Everglades Entrance Channel near Department Reference Monument R-85, in Broward County, with placement of beach compatible material to the south of the channel within the template at Dr. Von D. Mizell-Eula Johnson State Park (between R-86 and R-92) (**Figures 1 - 4**). The project also includes the following improvements to the inlet entrance (in and around R-85): 1) removal of the western portion of the rubble spoil shoal (**Figures 1, 4, and 5**); 2) construction of a 577-foot long by 16-foot wide rock rubble barrier at the eastern portion of the spoil shoal removal area (situated north to south) (**Figure 6**); and 3) extension of the north inlet entrance jetty approximately 160 feet towards the northeast (**Figures 2 and 5**). Placement of beach-compatible sand from the sediment trap is authorized by Permit No. 0289308-004-JC in the area just north of the north entrance jetty (**Figure 7**) following excavation (*i.e.*, initial construction), but not following maintenance dredging (*i.e.*, subsequent dredging of the sediment trap).

Excavation of the sand trap and clearing/removing rubble from the rubble shoal will result in direct impacts to 16.65 acres of unconsolidated rubble and boulder habitat with reef resources (**Figures 4, 5 and 8**). These direct impacts will be offset by 8.49 acres of mitigation, which will include the re-exposure of natural hardbottom by removal of rubble from rubble sand/habitat and from the rubble spoil shoal crest (**Figures 4 and 8**), enhancement and expansion of Broward County's Stony Coral Nursery Program, construction of a 0.2-acre boulder reef coral nursery in the nearshore area just south of the rock rubble barrier (offshore of R-85) (**Figure 9**), and construction of a 0.3-acre boulder reef coral nursery off Ft. Lauderdale (offshore of R-65), in approximately 40 feet of water (**Figure 10**). Scleractinian corals, octocorals, queen conch (*Strombus gigas*), and long-spined sea urchins (*Diadema antillarum*) will also be relocated from the direct impact area prior to initial construction of the sand trap to avoid/minimize impacts (**Figures 4 and 5**). Areas within the limits of the permitted turbidity mixing zone (500 m eastward and 300 m northward of the direct impact area and 1,000 m west along the north side of Port Everglades Inlet channel) are within the indirect impact area of the project (**Figure 8**). Impact avoidance measures (relocating federally listed scleractinian corals) will be employed in this area to minimize potential adverse effects from elevated turbidity during project construction. For additional information on benthic habitats in the project area and project impacts to hardbottom habitats see Sections 3.5 and 4.7 of the Environmental Assessment (CEG, 2015), the 2015 Benthic Habitat Mapping and Characterization Reports, and Benthic Resource Impact Summary Reports (NSUOC 2015a; 2015b).

This Plan is divided into six (6) main sections, four sections pertain directly to permit required minimization, mitigation, and monitoring (**Sections 2.0 – 5.0**), and two (2) sections described quality assurance/quality control and reporting procedures (**Sections 6.0 and 7.0**). The six (6) main sections of the plan are as follows: Minimization Efforts (**Section 2.0**), Sand Trap and Turbidity Mixing Zone Monitoring (**Section 3.0**), Mitigation Monitoring (**Section 4.0**), Beach Placement Site Monitoring (**Section 5.0**), Monitoring QA/QC and Schedule (**Section 6.0**), and Reporting (**Section 7.0**). All **Figures** cited in the report

are located in **Section 8.0**. **Appendices A - E** contain tables summarizing activities and monitoring requirements for **Sections 2.0, 3.1, 3.2, 4.0, and 5.0**.

2.0 IMPACT AVOIDANCE/MINIMIZATION

Impact avoidance/minimization required by FDEP and USACE permits entails the relocation of scleractinian coral and octocoral colonies, queen conch, and long-spined sea urchins to suitable locations outside the area potentially under the influence of the project. Relocation methods for corals located within the direct impact area (**Section 2.1.1**) and within the indirect impact area (**Section 2.2**) are described below, as are methods for relocating queen conch and long-spined sea urchins within the direct impact area (**Section 2.1.2**).

2.1 Direct Impact Area

For avoidance/minimization efforts, the direct impact area includes the footprint of the sediment trap, the rubble shoal removal area from where unconsolidated rubble will be cleared/removed, and a 10-m buffer around the rubble shoal removal area (**Figures 4 and 5**). Prior to construction, scleractinian corals 10 cm or more in diameter (major axis) and octocorals 15 cm or more in height will be relocated from the direct impact area along with queen conch and long-spined sea urchins (if urchins are present at the time of construction).

2.1.1 Scleractinian Corals and Octocorals

2.1.1.1 Locating and Collecting Coral Colonies

All hardbottom habitats within the direct impact area defined in **Section 2.1** (except for the Inlet Channel Wall) shall be included in the coral colony search and removal area. Within this area, a series of 50-m diameter plots will be used to define smaller, more manageable collection areas. Prior to in-water work, the project GIS database will be used to overlay 50-m diameter plots on the benthic habitat map to ensure the entire search area is covered. The center point for each plot shall be recorded and used to define each *in situ* collection site. At each collection site, divers will swim survey tapes 25 m out from the center point on north, east, south, and west compass headings to further subdivide each collection site into quadrants. A diver will be assigned to each quadrant.

At each site, divers will locate colonies in their quadrant that meet the requirement for relocation. Only scleractinian coral colonies that are visibly free of disease and have an area of continuous living tissue with a diameter (longest axis) equal to or greater than 10 cm or octocorals that have an area of continuous living tissue with a height equal to or greater than 15 cm are required to be collected and relocated. Although colonies of *Acropora cervicornis* were not observed in the direct impact area during reconnaissance surveys, all healthy colonies of *A. cervicornis* present during the search and relocation effort, regardless of size, shall be collected and relocated from the direct impact area.

Once located, divers will manually dislodge colonies from the substratum. Colonies on loose rubble that can

be easily carried by divers will be left on rubble. Colonies which are on colonized pavement or larger pieces of rubble will be removed from their substratum. Colonies shall be dislodged gently using hammers and chisels. Extreme care shall be taken to minimize impacts to colonies. Every effort shall be made to remove colonies from their substratum without fragmenting them. Some fragmentation is expected, particularly for colonies with flat growth forms. If a colony fragments during dislodgement, viable fragments shall be collected and re-attached at the receiver site.

Prior to removal, divers will note whether the colony is attached or not, and if attached, whether the colony is attached to rubble, consolidated pavement, or to large rubble/boulders. Once dislodged, divers will swim colonies to the center point of the site. Colonies that are removed from consolidated substrata will be collected separately from colonies on mobile rubble. After all quadrants at each site have been surveyed, the dislodged colonies will be lifted to the boat using buoyed baskets. Care shall be taken to minimize impact to colonies while in the baskets (e.g., colonies shall not be layered within the baskets). Colony collection shall occur in the beginning (morning) of each field day. Only as many colonies as can be transplanted the same field day shall be collected.

2.1.1.2 Transporting Collected Coral Colonies

Care shall be taken to prevent injury to colonies while transporting them to their respective transplantation sites. Prior to transport, each colony shall be wrapped in bubble wrap packing material to prevent abrasion, and then stowed in a cooler on board the vessel to keep them cool, damp, and out of the sun. Colonies shall only be kept in coolers (out of the water) for a few hours. Colonies shall be transported to their respective transplantation sites each field day following collection. The aim shall be to transplant all colonies the same day they were collected. If conditions (e.g., weather and sea state) do not allow for same day transplantation, then colonies will be transported to the transplantation site and cached in a safe, secure location. Attachment of the cached colonies shall be the priority for the next available field day.

2.1.1.3 Transplanting Collected Coral Colonies

2.1.1.3.1 Transplanting *Acropora cervicornis* Colonies

Although not expected within the direct impact area based on previous surveys, if colonies of *A. cervicornis* are encountered and collected within the direct impact area, then these colonies shall also be relocated to receiver sites A or B north of the project area, offshore of R-81 or R-76 in Broward County, respectively (**Figures 11a and 11b**). Water depths at the receiver site(s) to which colonies are relocated shall be similar to those in the project area from where they were collected (3-6 m).

Collected colonies shall be attached to the substratum using epoxy, nails, and zipties. Before attachment, the surface of the substratum will be prepared by removing fouling organisms (mostly algae) and sediment. Wire brushes and chisels will be used to prepare these surfaces. *Acropora cervicornis* colonies will be transplanted into circular plots, 7 m in diameter each. Each plot will have 10 to 15 colonies, and plots will be separated by 10 to 20 m.

2.1.1.3.2 Transplanting Non-Acroporid Scleractinian Colonies and Octocoral Colonies

All other coral colonies removed from the direct impact area shall be transplanted onto the Broward County Segment III mitigation boulder reef south of the project area, offshore of Hollywood Beach (R-102) (**Figure 12**). The mitigation boulder reef is located relatively close to the project area (approximately 5-6 km) in water depths (3-6 m) similar to those in the project area. Colonies will be attached to boulders using Portland cement; transplants will be attached to achieve a density of approximately 1.4 colonies/m². Before attachment, both the boulder and coral (underside of the colony or rubble) will be prepared. Fouling organisms (mostly algae) and sediment will be removed from contact surfaces so that the cement can form a solid bond between the colony and the boulder. Wire brushes and chisels will be used to prepare surfaces. Care will be taken to avoid injury to living coral tissue during surface preparation. Corals already in place on the boulder reef shall be avoided by transplanting the relocated corals to the “upper/top” surface of boulders.

2.1.1.4 Monitoring Transplanted Coral Colonies

2.1.1.4.1 Monitoring *Acropora cervicornis* Colonies

To document transplantation success, 25% of the transplanted plots will be monitored, which will be the equivalent of 25% of the colonies. Individual colonies will be tagged and each circular plot will also be tagged. All colonies contained within each tagged plot will be monitored. Because *A. cervicornis* naturally fragments, any fragments identified within plots during monitoring events will be included in the monitoring effort.

Acropora cervicornis colonies shall be monitored a total of 4 times: immediately following transplantation (baseline), and then at 3 months, 6 months, and 12 months (1 year) post-transplantation. During each monitoring event, colonies within each of the monitored plots will be measured (colony diameter) and have their condition recorded. The following shall be noted for colony condition: percent bleaching; percent old and recent mortality (respectively); and presence of disease, obvious signs of predation, and signs of sedimentation (burial of colony base etc.). The same monitoring procedures shall be used during each monitoring event (baseline through to Year 1). After each monitoring event, the status of the colonies within the plots will be compared to previous events and reported on.

2.1.1.4.2 Monitoring Non-Acroporid Scleractinian Colonies and Octocoral Colonies

To document the success of the transplantation process, 25% of the transplanted scleractinian corals and 25% of the transplanted octocorals shall be monitored. Each scleractinian coral species shall have at least 10 colonies that are monitored; species with 10 or fewer colonies shall have all their colonies monitored. Each octocoral genus shall have at least 100 colonies that are monitored; genera with 100 or fewer colonies shall have all their colonies monitored. Individual colonies will not be tagged; instead, individual boulders shall be tagged and all colonies on the boulders shall be mapped. All colonies on tagged boulders shall be monitored. The number of boulders included in the effort shall be sufficient to include the monitoring requirements stated above.

Octocoral and non-Acroporid scleractinian coral colonies shall be monitored a total of 4 times: immediately

following transplantation (baseline), and then at 6 months, 12 months (1 year), and 36 months (3 years) post-transplantation. During the baseline survey (immediately following transplantation), colonies on each of the monitored boulders shall be identified (to species for scleractinians, to genus for octocorals), measured (diameter for scleractinian corals, height for octocorals), and have their condition recorded. The following shall be noted for colony condition: percent bleaching; percent old and recent mortality; and presence of disease, obvious signs of predation, and sedimentation (accumulation). Whether a colony was relocated from rubble shall also be noted during the baseline survey. All colonies on tagged boulders shall be monitored over time, and the monitoring methods used in the baseline survey (identification, measurement of scleractinian coral diameter and octocoral height, recording of colony condition) shall be used in each subsequent survey. After each monitoring event, the status of the colonies (by species and genus, respectively), and any changes in status from previous events, will be discussed. Annual reports of survivorship will be submitted to the Department. Status updates in reports shall provide survival, condition, and tissue mortality data in tabular and graphical form.

2.1.2 Queen Conch and Long-Spined Sea Urchins

Direct impacts to queen conch (*Strombus gigas*) and long-spined sea urchins (*Diadema antillarum*) will be minimized/avoided by relocating individuals from the direct impact area immediately prior to dredging. Individuals of both species shall be collected in conjunction with coral collection at the 50-m diameter plots (see **Section 2.1.1.1**). Only adult urchins and mature queen conch will be collected; visually distinct morphological characteristics will be used to identify mature individuals of each species.

Mature queen conch shall be relocated to nearshore hardbottom offshore of (beyond) the permitted equilibrium toe of fill (ETOF) at Dr. Von D. Mizell-Eula Johnson State Park prior to project construction (**Figures 2 and 3**). Queen conch shall be collected in bins separate from coral colonies and transplanted to the receiver site on the same day as collection. Adult long-spined urchins shall be relocated to the *A. cervicornis* receiver sites (**Figures 11a and 11b**) or receiver sites for non-Acroporid scleractinian corals at the Broward County Segment III mitigation boulder reef (**Figure 12**). Urchins shall be collected in baskets and relocated on the day of collection. Urchins will be released to concentrate densities in specific receiver areas containing the coral transplants.

2.2 Indirect Impact Area

The permitted turbidity mixing zone for this project extends a maximum of 500 m eastward and 300 m northward of the direct impact area as well as 1,000 m west along the north side of Port Everglades Inlet channel (**Figure 8**). These areas are considered the indirect impact areas. Prior to construction of the initial project, colonies of *A. cervicornis*, and any other federally listed scleractinian species greater than or equal to 5 cm in diameter, shall be relocated from two (2) sites within the indirect impact area¹.

¹ Exhaustive surveys using the NMFS protocol documented colonies of *A. cervicornis* at sites 1, 2, 4, 33, and 34 in 2007 and 2014 (**Figure 13**). The number of colonies expected to be harvested from these sites was quantified in February 2017; an overall total of 1,962 colonies were counted at these five sites (**Figure 13**). Reconnaissance surveys conducted in December 2017 showed substantial declines in abundance, size, and colony condition following Hurricane Irma in September 2017. Overall abundance at the time of harvest is expected to be substantially less than the February 2017 counts. Coral colonies at sites 2, 33, and 34 will be monitored under federal requirements.

2.2.1 Locating and Collecting Scleractinian Coral Colonies

The colony search and removal area within the indirect impact area shall be confined to the two (2) sites (Sites 1 and 4) where previous exhaustive surveys of the indirect impact area (conducted using the National Marine Fisheries Service protocol) documented colonies of *A. cervicornis* in 2007, 2014, and 2017 (**Figure 13**). A 50-m diameter plot shall be established around the center point (location coordinates) of each of the sites. The area within each of the two 50-m diameter plots shall define the colony collection area for the indirect impact area. At each of the collection sites, divers will swim survey tapes 25 m out from the center point on north, east, south, and west compass headings to further subdivide each collection site into quadrants. A diver will be assigned to each quadrant.

At each site, divers will locate colonies of the federally-listed species, *Acropora cervicornis*, and any other federally listed species greater than or equal to 5 cm in diameter. Only colonies that are visibly free of disease will be collected and transplanted to receiver sites A or B (**Figures 11a and 11b**). Coral colony collection shall be conducted as described in **Section 2.1.1.1**.

2.2.2 Transporting Collected Scleractinian Colonies

Coral colony transportation shall be conducted as described in **Section 2.1.1.2**.

2.2.3 Transplanting Collected Scleractinian Colonies

All *A. cervicornis* colonies, and colonies of any other federally listed species greater than or equal to 5 cm in maximum dimension, collected within the indirect impact area shall be relocated to receiver sites A or B north of the project area (**Figures 11a and 11b**). The transplantation area to which they are relocated shall be of a similar depth (3-6 m) as the project area from which they were collected. Collected colonies shall be attached to the substratum using Portland cement, or by using epoxy, nails, and zipties. Before attachment, both the substratum and the coral (underside of the colony or rubble) will be prepared. Fouling organisms (mostly algae) and sediment will be removed from contact surfaces. Care will be taken to avoid injury to living coral tissue during surface preparation. Colonies will be transplanted into circular plots, 7 m in diameter each. Each plot will have 10 to 15 colonies, and plots will be separated by 10 to 20 m.

2.2.4 Monitoring Transplanted Scleractinian Colonies

To document the success of the transplantation process, 25% of the transplanted plots will be monitored, which will be the equivalent of 25% of the colonies. If there are 100 or fewer total colonies, all plots will be monitored. Each federally-listed scleractinian coral species with 10 or fewer colonies shall have all its colonies monitored. Individual colonies shall be tagged and each circular plot shall also be tagged. All colonies contained within each tagged plot shall be monitored. Any fragments of *A. cervicornis* observed within plots during monitoring events shall be included in the monitoring effort.

Colonies of *A. cervicornis* and any other federally-listed scleractinian species shall be monitored a total of 4 times: immediately following transplantation (baseline), and then at 3 months, 6 months, and 12 months (1 year) post-transplantation. During the baseline survey (immediately following transplantation),

colonies within each of the monitored plots shall be identified (to species), measured (diameter of maximum dimension), and have their condition recorded. The following shall be noted for colony condition: percent bleaching; percent old and recent mortality (respectively); and presence of disease, obvious signs of predation, and signs of sedimentation (burial of colony base etc.). Whether a colony was relocated from rubble shall also be noted during the baseline survey. All colonies within tagged monitoring plots shall be monitored over time, and the monitoring methods used in the baseline survey (identification, measurement of coral diameter, recording of colony condition) shall be used in each subsequent survey. After each monitoring event, the status of the colonies (by species), and any changes in status from previous events, will be discussed. Annual reports of survivorship will be submitted to the Department. Status updates in reports shall provide survival, condition, and tissue mortality data in tabular and graphical form.

3.0 SAND TRAP AND TURBIDITY MIXING ZONE AREA MONITORING

Construction area monitoring is divided into the following two main sections: 1) monitoring required for the initial excavation of the sand trap (**Section 3.1**); and, monitoring required for each sand trap maintenance dredging event (**Section 3.2**). Monitoring requirements for each of these events are provided below.

3.1 Sand Trap Excavation Event Monitoring

Sedimentation monitoring around and within the turbidity mixing zone for the excavation event/initial dredging event (500 m eastward and 300 m northward of the direct impact area and 1,000 m west along the north side of Port Everglades Inlet channel) (**Figure 8**) shall include a pre-construction (baseline) survey immediately prior to construction, during-construction surveys, and an immediate post-construction survey (within one [1] week of construction). The objective of these surveys is to document any unpermitted direct and/or secondary adverse impacts to nearshore hardbottom resources that may occur due to increased levels of sedimentation (sediment accumulation) resulting from the project. As such, during- and post-construction surveys will document changes in sediment depth and condition of biotic assemblages in relation to the excavation of the sand trap. Colonized pavement, ridge, and rubble/colonized pavement habitats within (**Section 3.1.1**) and adjacent to the turbidity mixing zone (**Section 3.1.2**) will be monitored using transects at permanent stations with additional during-construction turbidity measurements as triggers for sediment and benthic assessments (**Section 3.1.3**). If present, seagrass beds within the turbidity mixing zone will also be monitored to document any unpermitted impacts, should they occur (**Section 3.1.4**).

3.1.1 Monitoring of Permanent Sediment Stations within the Turbidity Mixing Zone

Biological monitoring of permanent stations within the turbidity mixing zone shall include a pre-construction (baseline) survey immediately prior to construction, weekly during-construction surveys, and an immediate post-construction survey (within one [1] week of construction). The pre-construction survey shall serve as the baseline for all subsequent excavation (initial sand trap construction) monitoring events. Monitoring shall consist of interval sediment depth measurements (pre-, during-, and post- construction), quantitative photo surveys (pre- and post- construction), scleractinian and octocoral surveys (pre- and post-construction), and qualitative video surveys (pre-construction, and possibly during- and post-construction [see **Section 3.1.1.6**]) as described in the following sections. During- and post-construction survey data shall

be compared to pre-construction (baseline) survey data to evaluate changes in benthic community cover/composition and size class distribution before/after dredging to determine whether project related impacts have occurred.

3.1.1.1 Monitoring Station Establishment

A network of 13 permanent monitoring stations shall be established within the turbidity mixing zone prior to construction. Prior studies in 2007, 2014, and 2015 sampled several sites in the project area; permanent stations shall be established at 13 of these sites (**Table 1 and Figure 14**). A single, permanent transect shall be established at each of the 13 permanent monitoring stations prior to construction.

Table 1. Permanent monitoring station transect length and direction.

Station No.	Transect Length (m)	Transect Direction (from – to)
1	150	S – N
2	50	N – S
4	50	W – E
5	50	W – E
22	50	W – E
28	50	S – N
31	50	S – N
34	50	N – S
35	50	S – N
36	50	E – W
38	150	S – N
40	50	W – E
45	50	N – S

See **Figure 14** for monitoring station and transect layout.

Permanent transects shall be of two lengths: 50 m long; and 150 m long. The 150-m transects shall only be established at monitoring stations 1 and 38; 50-m transects shall be established at all other stations (**Table 1**). The orientation (direction of travel from the start [meter 0] to the end of the transect) that each transect shall be established in is provided in **Table 1** (and see **Figure 14**). Transect orientations are station specific and relate to the position of the station with respect to the edge of the turbidity mixing zone. Once established, transects shall be permanent, and their locations and orientations shall not change in subsequent surveys (*i.e.*, during- and post-construction surveys). Permanent markers (pins, iron rods, etc.) will be installed at the start and end of each transect line to ensure repeatability in transect placement during all monitoring events. For each 50-m long transect (N=11 stations), an additional pin shall be installed in the middle of the transect at meter 25. For each 150-m transect (N=2 stations), additional pins shall be installed at meters 25, 50, 75, and 100.

The full length of each permanent transect shall be surveyed during the baseline monitoring event (see **Table 1**). The full length of each 50-m transect shall also be surveyed during each during-construction survey (weekly) and immediately following construction. However, unless during-construction monitoring indicates sedimentation impacts (see **Section 3.1.1.6**), only the first 50 m along the two 150-m transects (stations 1 and 38) shall be surveyed during each during-construction survey (weekly) and immediately

following construction.

3.1.1.2 Interval Sediment Depth Measurements

Sediment depth measurements will be conducted immediately prior to construction, weekly during construction, and immediately following construction to assess potential project-related sediment accumulation on hardbottom and/or biota. Unless during-construction monitoring indicates sedimentation impacts (see **Section 3.1.1.6**), only the first 50-m segment (0 – 50 m) along the two 150-m transects (stations 1 and 38) shall be surveyed during each during-construction survey (weekly) and immediately following construction. The full length (50 m) of all other transects shall be surveyed in their entirety during each monitoring event. See **Section 3.1.1.6** for sedimentation impact triggers for the 13 permanent monitoring stations.

During each monitoring event at each permanent station, sediment depth will be measured at 1-m intervals along each transect, inclusive of sand patches. Three (3) sediment depth measurements will be taken at each 1-m interval: one measurement on the center of the line, one 0.5 m to the left of the line, and another 0.5 m to the right of the line. For each measurement, a stainless-steel ruler graduated in millimeters (measuring at least 30 cm long) will be pressed through the sediment until the ruler reaches the surface of hard substratum or is totally immersed in sand. Depth measurements will be recorded to the nearest half centimeter (*e.g.*, 0 cm, 0.5 cm, 1 cm, 1.5 cm, etc.). Measurements greater than 30 cm will be recorded as “> 30 cm”.

3.1.1.3 Quantitative Photo Surveys

High-definition digital still photographs of a quality sufficient to allow for post-collection quantitative image analysis shall be taken immediately prior to and following construction to assess potential project-related sediment accumulation on hardbottom and/or biota. The baseline survey shall be conducted along the full length of each transect (**Table 1**). The full length of all 50 m transects (N=11) shall again be surveyed immediately following construction. However, unless during-construction interval sediment depth measurements (**Section 3.1.1.2**) indicate sedimentation impacts (see **Section 3.1.1.6**), only the first 50 meters (0 – 50 m) along the two 150-m transects (stations 1 and 38) shall be required to be surveyed immediately following construction. If no during-construction impacts are documented, then post-construction quantitative photo data collected along the first 50 m of transects at stations 1 and 38 shall be compared to pre-construction (baseline) quantitative photo data collected along the first 50 m of these transects.

During each monitoring event, a diver will photograph the full, required length of both sides of all 13 transects (see above). All photographs will be taken from a constant distance of 40 cm above the bottom. An aluminum bar attached to the camera housing (or other device) shall be used to maintain this distance. Post-collection image analysis shall include a quantitative assessment of functional group cover. Analysis shall be conducted using a point count method. A unique set of 25 random points shall be applied to each image. Functional group cover shall be assessed at each of the 25 points on every image. Percent cover of the following functional groups shall be assessed: sediment, bare substratum, macroalgae (total cover, and cover of Phaeophyta, Chlorophyta, and Rhodophyta), turf algae, cyanobacteria, encrusting calcareous algae,

sponges, scleractinian corals, octocorals, zoanthids, hydrocorals, hydroids, sessile annelid (not including *Phragmatopoma* spp.), wormrock (*Phragmatopoma* spp.), barnacle, bivalve, bryozoan, echinoderm, and tunicate. Each functional group will be assigned a percent cover value from 0 to 100%, with the total of all functional groups equaling 100%.

3.1.1.4 Coral and Octocoral Surveys Along Belt Transects

Immediately prior to and immediately following sand trap construction, a 50 m long by 1 m wide belt transect shall be used to survey scleractinian corals and octocorals at each of the 13 permanent stations. The permanent transect at each site shall form one side of the belt transect; for stations with a 150-m transect (stations 1 and 38), only the main, 50-m segment (0 – 50 m) shall be surveyed. The outer side of the belt transect shall fall 1 m from either the left or right of the permanent transect line. The side to which the belt transect falls (with respect to the transect line) at each site shall be recorded during the pre-construction survey and shall be repeated during the post-construction survey. During each survey, every scleractinian coral colony greater than 3 cm in maximum dimension within the belt transect shall be identified to species, and its size (maximum colony dimension) shall be measured to the nearest centimeter. Each octocoral within the belt transect shall be identified to genus, and its height shall be measured and assigned to one of the following four size classes: <5 cm, 5-25 cm, 25-50 cm, and >50 cm. Observed signs of sediment stress (e.g., accumulation on organisms or partial/total burial of organisms) shall be recorded and reported.

3.1.1.5 Qualitative Video Surveys

High definition digital video of a quality sufficient to allow for post-collection analysis shall be taken along the full length of all transects during the pre-construction (baseline) survey (see **Table 1**). Unless during-construction monitoring at these stations indicates sedimentation impacts, video surveys will not be conducted again. See **Section 3.1.1.6** below for triggers that would initiate during-construction and post-construction video surveys; this section also includes a trigger that would terminate during-construction video surveys (the post-construction survey shall still be required).

During each required survey, a diver will swim the full length of the transect at a speed of approximately 3.5 meters/minute with a constant camera distance of 40 cm above bottom. A convergent laser guidance system will be used to indicate the precise height of the camera at 40 cm from the bottom. If the diver is moved off the transect by waves or currents, they shall return to the point where they were disturbed and resume filming. The video shall be reviewed during the survey to ensure that no data gaps occurred from diver error and that the video quality is acceptable for analysis. Any missing video transect data or poor-quality video shall be re-filmed during the sampling event if possible.

3.1.1.6 Sedimentation Triggered Monitoring

Sedimentation impacts are defined as an increase in mean sediment depth of 1 cm or more above mean baseline sediment depth. If during-construction weekly sediment depth measurements document sedimentation impacts, then this shall trigger additional monitoring for impacted 50-m transect stations (**Section 3.1.1.6.1**) and for impacted 150-m transect stations (**Section 3.1.1.6.2**). If triggered, the monitoring

firm shall notify the JCP Compliance Officer as soon as possible that this trigger has been tripped.

3.1.1.6.1 Sedimentation Triggered Monitoring for 50-m Transects

If during-construction, weekly sediment depth measurements at any of the eleven (11) 50-m transect permanent stations (see **Table 1**) indicate sedimentation impacts, this will trigger the addition of qualitative video surveys in the weekly during-construction monitoring for the affected transect(s). The addition of qualitative video surveys to the weekly monitoring protocol shall be implemented the very next survey (next weekly during-construction survey) after sedimentation impacts are documented. Sediment triggered protocol changes shall remain in effect until either the post-construction survey is conducted, or weekly sediment depth data (at whichever station(s) were affected) indicates that impacts have receded/abated, which shall be defined as the return of mean sediment depth to that (or below that) recorded during the baseline survey.

3.1.1.6.2 Sedimentation Triggered Monitoring for 150-m Transects

If during-construction, weekly sediment depth measurements at stations 1 and/or 38 indicate sedimentation impacts along the main, 50-m segment of the transect, then changes to the weekly, during-construction monitoring protocol shall be triggered. The following changes shall be made if impacts to the main 50-m segment are detected: 1) interval sediment depth measurements shall be collected along the full, 150-m length of the transect; and 2) qualitative video surveys (see **Section 3.1.1.5** above) shall be initiated and shall be conducted along the full, 150-m length of the transect(s). Changes to the monitoring protocol shall be implemented during the next survey (next weekly during-construction survey) after sedimentation impacts are documented. Sediment triggered protocol changes shall remain in effect until either the post-construction survey is conducted, or weekly sediment depth data (at whichever station(s) are affected) indicate that impacts have receded/abated along the full 150-m length of the transect (*i.e.*, sediment depth measurements document a return of mean sediment depth to that (or below) that recorded along the first 50 m of the transect during the baseline survey).

3.1.2 Monitoring of Permanent Sediment Stations Adjacent to the Turbidity Mixing Zone

In addition to establishing and monitoring permanent sediment stations entirely within the turbidity mixing zone (**Section 3.1.1**), transects shall also be established and monitored in areas adjacent to (and slightly within) the turbidity mixing zone, at Stations 23 and 24 (**Figure 14** and see **Section 3.1.2.1.1**) and at the *Acropora cervicornis* monitoring station (indicated by the red star in **Figure 14**, and see **Section 3.1.2.1.2**). Surveys of permanent transects at these stations shall consist of interval sediment depth measurements [pre-, during-, and post- construction] and qualitative video (pre-construction and possibly during- and post-construction) as described below. See **Section 3.1.2.5** below for during- and post-construction monitoring triggers.

3.1.2.1 Monitoring Station Establishment

3.1.2.1.1 Additional Transect Establishment at Stations 23 and 24

Permanent, 70 m long, west-east oriented (shore perpendicular) transects are already established at stations 23 and 24 (pink transects at stations 23 and 24 in **Figure 14**). These transects, which were installed and are monitored in conjunction with the Broward County long-term nearshore hardbottom monitoring program, shall also be monitored for the Port Everglades Sand Bypassing project. At each of these stations, an additional, permanent, 100-m long transect shall be established prior to construction.

At biological monitoring station 23, the additional transect shall be oriented north to south. The north-south transect will start (zero [0] meter mark of the north-south transect) at the midpoint (meter mark 35) of the existing 70-m west-east transect and will extend 100 m to the south, towards the mixing zone. At biological monitoring station 24, the additional transect shall be oriented south to north. The south-north transect will start (zero [0] meter mark of south-north transect) at meter mark 25 of the existing 70-m west-east transect and will extend 100 m to the north, towards the northern edge of the mixing zone. To ensure repeatability in transect placement in case during- and post-construction monitoring events are triggered (see **Section 3.1.2.5** below), permanent markers (pins, iron rods, etc.) shall be installed at the start (meter zero [0]), end (meter 100), and at the 25, 50, and 75 meter marks of each of the new 100-m transects at stations 23 and 24.

3.1.2.1.2 Transect Establishment at the Segment II *Acropora cervicornis* Monitoring Station

Two (2) permanent transects, one running west to east and the other north to south, shall be established at the Broward County Segment II Nourishment project *Acropora cervicornis* monitoring station (indicated by the red star in **Figure 14**). The west-east transect line shall be 50 m in length, and the north-south transect line shall be 100 m in length. The origin (starting point [meter 0]) of the two transect lines shall be the 0 m mark (western end) of the west-east transect. The north-south transect will extend 100 m to the south, towards the mixing zone. To ensure repeatability in transect placement in case during- and post-construction monitoring events are triggered (see **Section 3.1.2.5** below), permanent markers (pins, iron rods, etc.) shall be installed at the start (zero [0] meter mark), end (50 meter mark), and in the middle (at the 25 meter mark) along the 50-m west-east transect and at the start (zero [0] meter mark), end (100 meter mark), and at the 25, 50, and 75 meter marks along the 100-m north-south transect at the *A. cervicornis* monitoring station.

3.1.2.2 Interval Sediment Depth Measurements

Immediately prior to construction (baseline), interval sediment depth measurement shall be collected (as described above, in **Section 3.1.1.2**) along the full length of all transects at stations 23, 24, and at the *A. cervicornis* monitoring station. Unless during-construction sediment depth monitoring indicates sedimentation impacts at stations 1 and/or 38 (see **Section 3.1.2.5** below), transects at stations 23, 24, and at the *A. cervicornis* monitoring station will not be required to be monitored again (*i.e.*, during- and/or post-construction interval sediment depth measurements will not be required), and the data collected during the baseline survey will not be analyzed.

3.1.2.3 Quantitative Photo Surveys

Immediately prior to construction (baseline), high definition digital still photographs shall be collected (as described above, in **Section 3.1.1.3**) along the full length of all transects at stations 23, 24, and at the *A. cervicornis* monitoring station. Unless during-construction sediment depth monitoring indicates sedimentation impacts at stations 1 and/or 38 (see **Section 3.1.2.5** below), transects at stations 23, 24, and at the *A. cervicornis* monitoring station will not be required to be monitored again (*i.e.*, post construction quantitative photo surveys will not be required), and photos collected during the baseline survey will not be analyzed.

3.1.2.4 Qualitative Video Surveys

High-definition digital video of a quality sufficient to allow for post-collection analysis shall be taken (as described above, in **Section 3.1.1.5**) along the full length of all transects at stations 23, 24, and at the *A. cervicornis* monitoring station prior to construction. Unless during-construction sediment depth monitoring indicates sedimentation impacts at stations 1 and/or 38 (see **Section 3.1.2.5** below), transects at stations 23, 24, and at the *A. cervicornis* monitoring station will not be required to be monitored again (*i.e.*, during- and/or post-construction qualitative video surveys will not be required).

3.1.2.5 Sedimentation Triggered Monitoring for Stations 23, 24, and *A. cervicornis*

The sedimentation impact trigger that shall initiate during- and post-construction monitoring of all transects at stations 23 and 24 and at the *A. cervicornis* monitoring station is tied to the sedimentation impact trigger for monitoring stations 1 and 38 (150-m transects) within the mixing zone (see **Section 3.1.1.6** above). If during construction, mean weekly sediment depth measurements at permanent sediment monitoring stations 1 and/or 38 indicate sedimentation impacts along the main, 50-m segment of the transect (*i.e.*, increase in mean sediment depth of 1 cm or more above mean baseline depth), then this shall trigger collection of interval sediment depth measurements along the full, 150-m length of the impacted transect(s) at these stations (1 and/or 38). If triggered, the monitoring firm shall notify the JCP Compliance Officer as soon as possible that this trigger has been tripped.

If, once a transect at either station 1 or 38 is extended (*i.e.*, monitoring of the entire 150 m transect length is triggered), weekly during-construction mean sediment depth measurements indicate sedimentation impacts along the distal, 100-m segment of the transect (from meter 50 to meter 150), then during- and post-construction monitoring of transects at stations 23 and 24 and at the *A. cervicornis* monitoring station shall be triggered. Sediment impact is defined as an increase in mean sediment depth of 1 cm or more above mean baseline sediment depth. If triggered, the monitoring firm shall notify the JCP Compliance Officer as soon as possible that this trigger has been tripped.

3.1.2.5.1 Sedimentation Triggered During-Construction Monitoring

If triggered (see **Section 3.1.2.5**), weekly during-construction monitoring of transects at stations 23 and 24 and at the *A. cervicornis* monitoring station shall commence the survey immediately following the survey in which sedimentation impacts were documented along a distal portion (meter 50 to meter 150) of a transect

at either station 1 or station 38. Weekly during-construction monitoring of transects at stations 23 and 24 and at the *A. cervicornis* monitoring station shall include interval sediment depth measurements (**Section 3.1.2.2** above) and qualitative video surveys (**Section 3.1.2.4** above) along the full lengths of all transects. During-construction survey data (sediment depth measurements and qualitative video) shall be compared to pre-construction survey data to evaluate changes in sediment and the benthic community to determine whether project-related impacts have occurred.

The requirement for weekly during-construction monitoring of all transects at stations 23 and 24 and at the *A. cervicornis* monitoring station shall remain in effect if during-construction mean sediment depth along any transect indicates sedimentation impacts (*i.e.*, increase in mean sediment depth of 1 cm or more above baseline mean sediment depth). Weekly during-construction monitoring of all transects at stations 23 and 24 and at the *A. cervicornis* monitoring station will be terminated if no sedimentation impacts are documented along any of the six (6) transects and weekly during-construction sediment depth data collected along the distal portions (from 50 to 150 m) of both transects at stations 1 and 38 indicate that sedimentation impacts have receded/abated. Recession/abatement of impacts shall be defined as the return of mean sediment depth along the distal portions of these transects to that (or below that) recorded during the baseline survey. Note that if the requirement for weekly during-construction monitoring of all transects at stations 23 and 24 and at the *A. cervicornis* monitoring station was terminated as described above, that subsequent sedimentation impacts along the distal portions (from 50 to 150 m) of transects at stations 1 and/or 38 will reinstate weekly during-construction monitoring of all transects at stations 23 and 24 and at the *A. cervicornis* monitoring station.

3.1.2.5.1 Sedimentation Triggered Post-Construction Monitoring

If during-construction monitoring of transects at stations 23 and 24 and at the *A. cervicornis* monitoring station is triggered, then post-construction monitoring is mandatory, and shall be conducted regardless of whether during-construction monitoring is terminated (see above). Post-construction surveys along the full length of all transects shall include interval sediment depth measurements (**Section 3.1.2.2**), quantitative photo surveys (**Section 3.1.2.3**), and qualitative video surveys (**Section 3.1.2.4**). All survey data (pre-, during-, and post-construction) shall be analyzed and during- and post-construction survey data shall be compared to pre-construction (baseline) survey data collected along transects at stations 23 and 24 and at the *A. cervicornis* monitoring station to evaluate changes in the depths and distribution of sediments and changes in benthic community cover/composition before/after dredging to determine whether project related impacts have occurred.

3.1.3 Additional Turbidity Monitoring

Twice daily turbidity monitoring at the edge of the mixing zone is required by permit to ensure compliance with state water quality standards. Along with this typical water quality monitoring, additional twice daily turbidity monitoring will be conducted over the colonized pavement (NE of sand trap) and ridge habitats within the turbidity mixing zone north of Port Everglades Inlet (**Figure 4**). The purpose of this additional monitoring is to assess potential sediment deposition in consolidated hardbottom habitats outside of the permanent monitoring sites but within the mixing zone due to extended periods of elevated turbidity.

3.1.3.1 Turbidity Sample Collection

Additional turbidity samples will be collected two (2) times per day, at two (2) locations north of Port Everglades Inlet. Samples will be collected where the densest portion of the turbidity plume occurs over the north colonized pavement (NE of sand trap) and ridge habitats (**Figure 4**). Prior to construction, the Permittee shall provide the turbidity monitoring contractor with a shape file of the location and limits of the north colonized pavement and ridge habitat areas within the turbidity mixing zone. If the densest portion of the visible turbidity plume extends over these habitats during construction, then a single turbidity sample shall be collected at mid-depth. The Differential GPS location at which the sample is taken shall be recorded and provided along with the data to the Department.

3.1.3.2 Turbidity GIS Database and Turbidity Monitoring Trigger

A Turbidity Monitoring GIS database shall be maintained by the Biological Monitoring Consultant. When collected, the location of each additional turbidity sample shall be transmitted from the turbidity monitoring contractor to the Biological Monitoring Consultant within 18 hours of sample collection. Using these data in conjunction with the benthic habitat map, the Biological Monitoring Consultant shall track the daily locations and turbidity values in ArcGIS. If the densest portion of the plume extends within a 10-m radius of the same location for three (3) consecutive days, and if the turbidity level at the location exceeds 10 NTUs at each reading, then the benthic community at this location shall be inspected for sedimentation impacts on the following day. If triggered, the monitoring firm shall notify the JCP Compliance Officer as soon as possible that this trigger has been tripped. If adverse weather conditions prohibit inspection the following day, then the inspection will be conducted as soon as conditions allow for safe dive operations. If the location of elevated turbidity is within 50 m of a permanent monitoring site, then inspection of the site will be conducted in conjunction with the next weekly monitoring event at the permanent monitoring site. If weekly monitoring has already occurred that week at the permanent site, then the next weekly monitoring event for the permanent monitoring site shall be moved up (rescheduled) so that it occurs within 24 to 48 hours.

3.1.3.3 Sedimentation Inspection

A temporary 20 m long by 1 m wide belt transect shall be investigated at the location where elevated turbidity levels are recorded. If divers observe sediment deposition resulting in the burial (partial or total) of hardbottom biota during the inspection, then the transect shall immediately be permanently installed at the site (*i.e.*, permanent markers (pins, iron rods, etc.) shall be installed at the start, end, and in the middle [10-meter mark] of the transect to ensure repeatability of transect placement). Following installation, the area of impact (where burial is observed) shall be mapped and quantified. Digital still images shall be taken to document the current condition (impact) of organisms. Impacted sites shall be added to the list of permanent monitoring sites that are being monitored weekly during construction (see **Sections 3.1.1.1** and **3.1.2** for permanent monitoring sites). As with all other permanent monitoring sites within the turbidity mixing zone (all stations in **Table 1**), any impact sites that are added to the list of permanent monitoring sites shall be visited weekly during construction to determine the permanence of the impact (*i.e.*, success or failure of impacts to recede). The requirement for weekly during-construction (and immediate post-construction)

monitoring of an impact site shall remain in effect until either the post-construction survey is conducted, or weekly during-construction monitoring indicates impacts are no longer present.

3.1.3.4 Adaptive Management of Dredging Operations

Modification of dredging operations (*e.g.*, moving the dredge to another location in the sand trap or reducing the dredge production rate) shall be considered by the Permittee if it seems possible that such actions could potentially reduce the level of sediment deposition on the colonized pavement habitat. Such measures would be considered minimization efforts, as they could help avoid complete mortality and permanent impacts to hardbottom biota within the mixing zone.

3.1.4 Seagrass Monitoring

A few small, dense beds and patches of *Halophila decipiens* have historically been present within the turbidity mixing zone, along the north side of the inlet (**Figures 8 and 15**). Turbidity may be elevated over these beds during excavation (initial construction dredging) of the sand trap and removal of rubble from the rubble shoal, reducing the amount of light and possibly resulting in sedimentation on seagrasses. If seagrasses are present in this area immediately prior to excavation of the sand trap, then these beds shall be monitored immediately prior to construction, during-construction, and immediately following construction as described below.

3.1.4.1 Seagrass Bed Mapping

The edge (boundary) of each seagrass bed shall be mapped (delineated) immediately prior to and immediately following construction. Seagrass bed edge mapping shall also be conducted during construction if the turbidity trigger is tripped (see **Section 3.1.4.2** below). During each *in situ* delineation, divers shall swim around the full extent of each seagrass bed while towing a buoy equipped with a GNSS receiver which wirelessly transmits corrected positions to a radio on the survey vessel, allowing divers to work untethered to the vessel. The receiving radio is interfaced with ArcPad or HYPACK navigation software on the survey vessel. Delineation shall be used to determine the spatial arrangement and sizes (area) of each seagrass beds prior to, during- (if required), and following construction. The pre-construction edge of each bed shall be compared to the post-construction survey edge of each bed to determine whether impacts have occurred (*i.e.*, loss of spatial coverage).

3.1.4.2 Turbidity Monitoring Trigger

Turbidity shall be monitored weekly over the two largest beds of seagrass during sand trap dredging activities. For these seagrass beds, turbidity monitoring events shall be conducted on an incoming tide. If a turbidity reading at either of these locations exceeds 6 NTUs above background, then seagrass beds shall be visually inspected and mapped by marine scientists within 48 hours of the elevated reading. If adverse weather conditions or conflicts with Port traffic are encountered that would put the dive crew at risk, then monitoring will be conducted as soon as conditions allow for safe dive operations. If triggered, the monitoring firm shall notify the JCP Compliance Officer as soon as possible that this trigger has been tripped

If no adverse effects to seagrasses are discovered after three, separate during-construction surveys (visual inspections and mapping efforts) following exceedances of the 6 NTU above background trigger, then the *in situ* during-construction seagrass monitoring will be discontinued. Future *in situ* during-construction seagrass monitoring will only be required if weekly turbidity monitoring values at the seagrass beds indicate turbidity levels exceed state water quality standards of 29 NTUs above background. Post-construction mapping of the edge (boundary) of each seagrass bed will still be required even if *in situ* during-construction seagrass monitoring is discontinued during-construction.

3.1.4.3 Trigger Required Seagrass Bed Visual Inspection and Mapping

3.1.4.3.1 Visual Inspection

Scientists shall visually assess the two beds within each mixing zone by swimming an east-west transect through the extent of each bed. Seagrasses will be inspected for signs of sediment accumulation on above-ground biomass and partial/complete burial of patches due to sediment deposition.

If impacts are detected, then the seagrass site shall be added to the list of permanent monitoring sites that are monitored *in situ* during construction (see **Sections 3.1.1.1** and **3.1.2** for permanent monitoring sites). As with all other permanent monitoring sites within the turbidity mixing zone, any impact sites that are added to the list of permanent monitoring sites shall be visited weekly during construction to determine the permanence of the impact (*i.e.*, success or failure of impacts to recede). Once triggered, the requirement for weekly during-construction (and immediate post-construction) monitoring of an impact site shall remain in effect until either the post-construction survey is conducted, or weekly during-construction monitoring indicates impacts are no longer present.

3.1.4.3.2 Mapping

The edges (boundaries) of seagrass beds shall also be mapped in conjunction with the visual inspection (see **Section A3.1.4.3.1** above for methods). The edge of each bed mapped during the inspection survey shall be compared to the pre-construction survey edge of each bed to determine whether impacts have occurred (*i.e.*, loss of area or change in distribution). If impacts are detected, then maps shall be used to determine and quantify the area of loss or degradation (impact). Post-construction mapping of the edge of each seagrass bed shall be used to determine the permanence of any impacts documented during construction.

3.2 Maintenance Dredging Event Monitoring

Maintenance dredging of the excavated sand trap is expected to occur at least once every 2 to 4 years. The permitted turbidity mixing zone for maintenance dredging events extends a maximum of 150 m from the sand trap (**Figure 16**). Based on the expected post-project condition from exposure of rubble/colonized pavement following sand trap excavation and removal of unconsolidated rubble, there is expected to be approximately 1.1 acres of colonized pavement and 8.2 acres of rubble/colonized pavement within the 150-m mixing zone around the sediment trap. These hardbottom areas within the 150-m turbidity mixing zone are considered indirect impact areas for maintenance dredging events and will be monitored using transects at newly-established permanent monitoring stations (*i.e.*, transects at stations that are different

from the permanent stations monitored for sand trap excavation).

3.2.1 Permanent Sediment Station Monitoring

Biological monitoring of permanent stations within the 150-m maintenance dredging turbidity mixing zone shall include a pre-construction (baseline) survey immediately prior to construction, weekly during-construction surveys, and an immediate post-construction survey (within one [1] week of construction). Monitoring surveys shall consist of sediment depth measurements, collection and analysis of quantitative photographs, collection of qualitative video, and identification and measurement of scleractinian and octocorals as described in the following sections. The aim of surveys conducted during each monitoring event is to document any unpermitted direct and/or secondary adverse impacts to nearshore hardbottom resources that may occur due to increased levels of sedimentation (sediment accumulation) resulting from the project. As such, during- and post- construction survey data shall be compared to pre-construction survey data to evaluate changes in sediment depth and distribution as well as changes in benthic community cover/composition and size class distribution before/after dredging to determine whether project related impacts have occurred. Note that **Section 3.2.1.6** below provides information on sedimentation impacts that would trigger changes (additional surveys) to the weekly during-construction monitoring protocol and that **Section 3.2.1.7** provides information on the trigger that would terminate during-construction maintenance dredging monitoring (post-construction monitoring would still be required if this trigger were tripped).

3.2.1.1 Monitoring Station Establishment

A network of six (6) permanent monitoring stations shall be established in the colonized pavement and rubble/colonized pavement habitats within the 150-m mixing zone around the sediment trap. The locations of the six (6) permanent stations shall be determined in consultation with Department and USACE staff prior to the first maintenance dredging event. Locations will be based on post-project habitat conditions and Department staff assessment. Once determined, the Minimization, Mitigation, and Monitoring Plan shall be updated to include the coordinates of the six (6) new permanent stations and to incorporate a map of their positions. Once established, the same monitoring stations and transects (same location, orientation, and length) shall be monitored during each successive maintenance dredging event.

A single, permanent, 50-m long transect shall be established at each monitoring station prior to the first maintenance dredging event. Once established, these transects shall be monitored in full during all subsequent surveys. Permanent markers (pins, iron rods, etc.) shall be installed at the start, end, and in the middle (25-m mark) of each transect to ensure repeatability of transect placement during all monitoring events. Monitoring station transect surveys will consist of interval sediment depth measurements (pre-, during-, and post- construction), quantitative photography with functional group analysis (pre- and post-construction), and coral colony identification and size measurements (pre- and post- construction) as described below.

3.2.1.2 Interval Sediment Depth Measurements

For each maintenance dredging event, sediment depth data shall be collected immediately prior to

construction, weekly during construction, and immediately following construction to assess potential project related sediment accumulation on hardbottom or hardbottom biota. During each monitoring event, sediment depth will be measured at 1-m intervals along the entire length of each transect at all six stations, inclusive of sand patches. Three (3) sediment depth measurements will be taken at each 1-m interval, one on the center line and one on either side of the transect, 0.5 m to the left and right of the center line, respectively. For each measurement, a stainless-steel ruler graduated in millimeters (measuring at least 30 cm long) will be pressed through the sediment until the ruler reaches the surface of hard substrata, or it is totally immersed in sand. Depth measurements will be recorded to the nearest half centimeter (*e.g.*, 0.0 cm, 0.5 cm, 1.0 cm, 1.5 cm, etc.). Measurements greater than 30 cm will be recorded as “> 30 cm”. **Section 3.2.1.6** below provides information on sedimentation impacts that, if documented, would trigger changes (additional surveys) to the weekly during-construction monitoring protocol. **Section 3.2.1.7** below provides information on the trigger that would terminate during-construction monitoring (*i.e.*, sediment depth measurements) if tripped.

3.2.1.3 Quantitative Photo Surveys

For each maintenance dredging event, high-definition digital still photographs of a quality sufficient to allow for post-collection analysis shall be taken immediately prior to and following construction. During each monitoring event, a diver will photograph the full length of both sides of each transect. All photographs will be taken from a constant distance of 40 cm above the bottom. An aluminum bar attached to the camera housing (or other device) shall be used to maintain this distance.

Post-collection image analysis shall include a quantitative assessment of functional group cover. Analysis shall be conducted using a point count method. A unique set of 25 random points shall be applied to each image. Functional group cover shall be assessed at each of the 25 points on every image. Percent cover of the following functional groups shall be assessed: sediment, bare substratum, macroalgae (total cover, and cover of Phaeophyta, Chlorophyta, and Rhodophyta), turf algae, cyanobacteria, encrusting calcareous algae, sponges, scleractinian corals, octocorals, zoanthids, hydrocorals, hydroids, sessile annelid (not including *Phragmatopoma* spp.), wormrock, barnacle, bivalve, bryozoan, echinoderm, and tunicate. Each functional group will be assigned a percent cover value from 0 to 100%, with the total of all functional groups equaling 100%.

3.2.1.4 Coral and Octocoral Surveys Along Belt Transects

Immediately prior to and immediately following each maintenance dredging event, a 50 m long by 1 m wide belt transect shall be used to survey scleractinian corals and octocorals at each of the six (6) permanent stations. The permanent transect at each station shall form one side of the belt transect. The other side of the belt transect shall fall 1 m away, to either the left or right of the permanent transect. The side to which the belt transect falls (with respect to the transect line) at each site shall be recorded during the pre-construction survey and shall be repeated during the post-construction survey. During each survey, every scleractinian coral colony greater than 3 cm in maximum diameter within the belt transect shall be identified to species, and its size (maximum colony dimension) shall be measured to the nearest centimeter. Each octocoral within the belt transect shall be identified to genus, and its height shall be measured and assigned to one of

the following four size classes: <5 cm, 5-25 cm, 25-50 cm, and >50 cm. Observed signs of sediment stress (e.g., accumulation on organisms or partial/total burial of organisms) shall be recorded and reported.

3.2.1.5 Qualitative Video Surveys

High-definition digital video of a quality sufficient to allow for post-collection analysis shall be collected at all six (6) monitoring stations immediately prior to each maintenance dredging event. Unless during-construction monitoring at these stations indicates sedimentation impacts, video surveys will not be conducted during or following construction. See **Section 3.1.2.6** below for triggers that would initiate during- and post-construction video surveys; this section also includes a trigger that would terminate during-construction video surveys. If during-construction video surveys are triggered, then a post-construction video survey shall also be conducted.

During each required survey, a diver will swim the full length of the transect at a speed of approximately 3.5 meters/minute with a constant camera distance of 40 cm above bottom. A convergent laser guidance system will be used to indicate the precise height of the camera at 40 cm from the bottom. If the diver is moved off the transect by waves or currents, they shall return to the point where they were disturbed and resume filming. The video shall be reviewed during the survey to ensure that no data gaps occurred from diver error and that the video quality is acceptable for analysis. Any missing video transect data or poor-quality video shall be re-filmed during the sampling event.

3.2.1.6 Sedimentation Triggered Monitoring

If, during construction, weekly sediment depth measurements at any of six (6) permanent stations indicate sedimentation impacts, then this will trigger the addition of qualitative video surveys in the weekly during-construction monitoring for the affected transect(s). Sediment impacts are defined as an increase in mean sediment depth of 1 cm or more above mean baseline sediment depth. If triggered, the monitoring firm shall notify the JCP Compliance Officer as soon as possible that this trigger has been tripped. The addition of qualitative video surveys to the weekly monitoring protocol shall be implemented the very next survey (next weekly during-construction survey) after sedimentation impacts were documented. Sediment triggered protocol changes shall remain in effect until either the post-construction survey is conducted, or weekly sediment depth data (at whichever station(s) were affected) indicates that impacts have receded/abated, which shall be defined as the return of mean sediment depth to that (or below that) recorded during the baseline survey.

3.2.1.7 Triggers for Terminating and Reinstating During-Construction Monitoring

Pre-construction and post-construction survey data shall be compared to evaluate changes in benthic community cover/composition and size class distribution before/after maintenance dredging to evaluate changes in the hardbottom community. If significant sediment deposition exceeding 2 cm in depth over colonized pavement and rubble/colonized pavement habitats is not recorded at any of the six (6) monitoring stations after two separate maintenance dredging events, then during-construction maintenance dredging turbidity area monitoring shall be discontinued. Pre- and post- construction monitoring, as

described in **3.2.1.2 – 3.2.1.4** above (*i.e.*, interval sediment depth measurements, quantitative photos surveys, and coral identification and size measurement), shall still be required for each maintenance dredging event.

If, after the trigger for terminating during-construction monitoring has been tripped, post-construction survey data collected for a maintenance dredging event indicates that significant sediment deposition exceeding 2 cm in depth over colonized pavement and rubble/colonized pavement habitats has occurred, then during-construction monitoring shall be reinstated for future maintenance dredging events. The trigger described in this section (for termination of during-construction monitoring) would again apply if during-construction monitoring were reinstated (*i.e.*, if, after during-construction monitoring is reinstated, no significant sediment deposition exceeding 2 cm in depth over colonized pavement and rubble/colonized pavement habitats is documented after two further, separate maintenance dredging events, then during-construction monitoring would again be terminated).

4.0 MITIGATION MONITORING

Methods of monitoring and success criteria for the following three permit required mitigation activities are provided in **Sections 4.1, 4.2, and 4.3** below, respectively.

- Re-exposure of colonized pavement from partial removal of the rubble shoal and shoal crest;
- Placement of artificial reef material (limestone boulders) to create new coral nursery habitat; and
- Relocation of corals of opportunity to the new coral reef nursery habitat (artificial reefs) and active management of the new habitat to enhance and expand the Broward County coral nursery program.

4.1 Re-exposure of Hardbottom Mitigation

Removal of roughly 16.2 acres of highly mobile, unconsolidated rubble from rubble sand/habitat and from the rubble spoil shoal crest during initial excavation of the sand trap is expected to re-expose approximately 7.3 acres of colonized pavement habitat: 4.55 acres of habitat from the direct removal of rubble from the Rubble/Sand habitat; and 2.76 acres from partial removal of the rubble shoal crest (**Figure 5**). Placement of a limestone boulder retaining wall will reduce landward movement of rubble into the project area after trap construction (**Figure 6** and see the 5.5 m (18 ft) depth contour (red line) in **Figure 17**). Because a minor rubble component will remain after clearing activities, the post-project habitat is expected to provide functions similar to the existing rubble/colonized pavement habitat in the project area.

4.1.1 Monitoring of Re-exposed Hardbottom

Post-construction monitoring shall document the exposed acreage (**Section 4.1.1.1**) and ecological functions (**Section 4.1.1.2**) of the re-exposed rubble/colonized pavement habitat in the project area. Habitat maps shall be created (**Section 4.1.1.1**) and *in situ* benthic surveys of sites within the re-exposed direct impact hardbottom area and of reference community sites shall be conducted (**Section 4.1.1.2**). Monitoring methods for these different tasks are provided in the following sections. **Section 4.1.2** identifies triggers that would terminate further post-construction monitoring. Mitigation success criteria for this effort are set forth

in **Section 4.1.3**.

4.1.1.1 Benthic Habitat Maps

Post-construction benthic habitat maps that document the acreage of re-exposed hardbottom shall be created at Years 1, 2, 3 and 5 post-construction (but see the monitoring termination trigger in **Section 4.1.2**). Benthic habitat maps shall be based on desk-top analysis (**Section 4.1.1.1.1**) and diver mapping and ground-truthing of the hardbottom edge/habitat borders (**Section 4.1.1.1.2**), as described below.

4.1.1.1.1 Desktop Work

The purpose of this task is to document the area (acreage) of re-exposed hardbottom at set post-construction time points by updating the 2015 pre-construction benthic habitat map through a desktop analysis of current (Years 1, 2, 3, and 5 post-construction) high-resolution, georectified aerial photographs and comparison to the existing habitat delineations using ArcGIS software. Analysis will focus on the borders between habitat types in the study area (Shoal Crest, Rubble/Sand, Colonized Pavement, Channel Wall, Boulders, and Unconsolidated Sediments). For each year's post-construction map, habitats which visually appear distinctly different from the 2014/15 condition will be modified according to current post-project aerial photographs and further refined based on field reconnaissance surveys (**Section 4.1.1.1.2**). The 2015 benthic habitat polygons will be visually inspected with respect to the new data and new polygon edges will be drawn in GIS where necessary to update the map. The acreage of re-exposed rubble/colonized pavement habitat will be calculated based on updated maps at each required post-construction time point.

4.1.1.1.2 *In Situ* Delineation of Hardbottom Edge and Habitat Borders

Field reconnaissance will be performed to qualitatively validate (ground truth) the distinct habitats noted in the GIS visual interpretation (**Section 4.1.1.1.1**) and to aid in establishing habitat boundaries for map refinement. Reconnaissance surveys shall be conducted in conjunction with aerial photograph analysis (*i.e.*, at Years 1, 2, 3, and 5 post-construction). The actual habitat for each area in question will be qualitatively characterized *in situ* by research divers. Researchers will also note obvious habitat transitions while swimming predetermined courses spanning multiple habitats. Several habitat borders and boulder piles will be mapped by snorkel and GPS from the water surface. This method has proved to be successful in mapping detailed shallow-water habitats (Walker et al. 2012). GPS positions will be recorded by a snorkeler at the surface while swimming the habitat boundaries by towing a DPGS wireless receiver which transmits signals to the receiver on the survey vessel.

4.1.1.2 Benthic Surveys

Benthic surveys of re-exposed rubble/colonized pavement habitat and reference community rubble/colonized pavement habitat shall be conducted at Years 3 and 5 post-construction (but see the monitoring termination trigger in **Section 4.1.2**). Comparison of survey results between these two areas (re-exposed habitat vs. reference habitat) will be used to determine whether the re-exposed habitat is providing functions similar to the existing rubble/colonized pavement (reference) habitat.

A total of 16 sites shall be surveyed during each monitoring event. Eight (8) permanent sites shall be established within the re-exposed rubble/colonized pavement habitat; another eight (8) sites shall be established within rubble/colonized pavement habitat. Due to the limited acreage of reference community habitat, the eight (8) reference community sites may be located in areas within the turbidity mixing zone permitted for the sand trap excavation event and also for sand trap maintenance dredging events. The locations of all 16 sites shall be determined in consultation with Department staff prior to the initial (Year 3 post-construction) monitoring event. Locations will be based on post project habitat conditions and Department staff assessment. The Biological Monitoring Plan shall be updated to include the coordinates and a map of the positions of all 16 sites. Once established, the same monitoring sites, including transects (same locations and lengths) shall be monitored during each additional monitoring event (*i.e.*, Year 5 post-construction, if required).

In situ benthic monitoring shall be based on surveys conducted along permanent transects installed at each site. Transect monitoring shall consist of quantitative photo surveys with functional group cover analysis and quadrat/belt transect surveys.

4.1.1.2.1 Monitoring Station Establishment

A total of three (3) parallel 20-m long transects spaced 10 m apart shall be established at each site (N=16 sites) during the initial, Year 3 post-construction survey. Once established, these transects shall be monitored in full during all subsequent surveys (*e.g.*, Year 5 post-construction); the positions and lengths of transects shall not change in subsequent surveys. Permanent markers (pins, iron rods, etc.) shall be installed at the start, end, and in the middle (10 m mark) of each of the three transect lines at each site to and GPS coordinates for each permanent shall be recorded to ensure repeatability in transect placement during all monitoring events.

4.1.1.2.2 Quantitative Photo Surveys

High-definition digital still photographs of a quality sufficient to allow for post-collection analysis shall be taken along each of the three (3) 20 m transects at each site. During each survey, a diver shall photograph the full length of one (1) side of each transect; the side on which photographs are taken shall be recorded so that the survey may be repeated if subsequent monitoring events are necessary. All photographs shall be taken from a constant distance of 40 cm above the bottom. An aluminum bar attached to the camera housing (or other device) shall be used to maintain this distance.

Post-collection image analysis shall include a quantitative assessment of functional group cover. Analysis shall be conducted using a point count method. A unique set of 25 random points shall be applied to each image. Functional group cover shall be assessed at each of the 25 points on every image. Percent cover of the following 19 functional groups shall be assessed: sediment, bare substratum, macroalgae (total cover, and cover of Phaeophyta, Chlorophyta, and Rhodophyta), turf algae, cyanobacteria, encrusting calcareous algae, sponges, scleractinian corals, octocorals, zoanthids, hydrocorals, hydroids, sessile annelid (not including *Phragmatopoma* spp.), wormrock, barnacle, bivalve, bryozoan, echinoderm, and tunicate. Each

functional group will be assigned a percent cover value from 0 to 100%, with the total of all functional groups equaling 100%.

4.1.1.2.3 Quadrat/Belt Transect Survey

A quadrat/belt transect survey shall be used to document scleractinian coral species richness, colony size, density, and condition (presence of disease, bleaching, and *Cliona* spp.), as well as octocoral density and colony size at each site. For each monitoring event, 0.75 m² quadrats (0.75 m wide by 1 m long) shall be sampled along both sides of the middle 20 m transect at each site (N=20 quadrats per side). Sampling in this fashion will result in a total survey area of 30 m² (0.75 m + 0.75 m wide X 20 m long) per site. During the survey, every scleractinian coral colony within each quadrat shall be identified to species, and its size (maximum colony dimension) shall be measured to the nearest centimeter. Each octocoral within each quadrat shall be identified to genus, and its height shall be measured and assigned to one of the following four size classes: <5 cm, 5-25 cm, 25-50 cm, and >50 cm.

4.1.2 Termination of Post-Construction Monitoring Triggers

If, prior to Year 5 post-construction (e.g., Year 2 or 3 post-construction), results of the post-construction mapping effort (**Section 4.1.1.1**) indicate that the acreage of rubble/colonized pavement in the cleared impact area meets the acreage criterion set forth in **Section 4.1.3** (Criterion A), then the Permittee and/or authorized agent may request termination of this aspect of monitoring. If results of the Year 3 post-construction benthic survey (**Section 4.1.1.2**) indicate that the ecological functions and biological community in the re-exposed habitat meets the success criterion set forth in **Section 4.1.3** (Criterion B), then the Permittee and/or authorized agent may request termination of this aspect of monitoring. If both success criteria set forth in **Section 4.1.3** (Criteria A and B) have been achieved prior to the Year 5 monitoring event, then the Permittee and/or authorized agent may request termination of this portion of mitigation monitoring and that the regulatory agencies consider this aspect of the mitigation effort to be complete.

4.1.3 Re-exposure of Hardbottom Mitigation Success Criteria

By Year 5 post-construction (initial excavation) of the sand trap or earlier, the following success criteria shall be met:

- A. The acreage of rubble/colonized pavement in the cleared impact area shall be at least 90% of the predicted 7.3 acres of exposure (i.e., 6.6 acres or more shall be re-exposed).
- B. The major functional groups of the rubble/colonized pavement in the cleared impact area shall group at the 75% similarity level with rubble/colonized pavement reference sites located outside of the mitigation area.

If after the Year 5 survey, all mitigation criteria (A and B above) have not been met, then the Permittee and/or authorized agent shall meet with Department staff to discuss the performance of the mitigation and a path forward to complete the mitigation requirement. Additional mitigation may be required to fulfill the mitigation requirement.

4.2 Boulder Reef Scleractinian Coral Nursery Mitigation

To offset direct impacts due to construction of the project, two (2) coral nursery artificial reefs (one nearshore and one offshore boulder reef) totaling 0.50 acres in net area shall be constructed to expand and enhance the Broward County coral nursery program. These artificial reefs will provide stable substratum for 3,000 “corals of opportunity” that will be relocated to the mitigation reefs (see **Section 4.3**). Each mitigation boulder reef shall be surveyed once following construction (see **Section 4.2.2**) to provide the Department with reasonable assurance that the Permit required net acreage has been constructed (see success criteria in **Section 4.2.3**).

4.2.1 Coral Nursery Boulder Reefs

4.2.1.1 Nearshore Coral Nursery Boulder Reef

A 0.20-acre (net acreage) limestone boulder reef shall be constructed in the nearshore waters of the project area (near R-85) just south of the rubble barrier (**Figure 9**). The nearshore coral nursery boulder reef shall be located immediately seaward of the 150-m mixing zone for maintenance dredging and shall not be affected by elevated turbidity. This location is ideal due to its proximity to a known source of corals of opportunity (*i.e.* corals on unconsolidated rubble on the portion of the spoil shoal seaward of the rubble removal area). The nearshore coral artificial reef shall be constructed in conjunction with the limestone rubble barrier during the initial phase of project construction, and the limestone boulders in the direct impact area will be used to construct these structures if conditions allow for re-location of the boulders at the time of sand trap construction.

4.2.1.2 Offshore Coral Nursery Boulder Reef

Within one year of sand trap construction (initial excavation), a 0.30-acre (net acreage) limestone boulder reef shall be constructed offshore of Fort Lauderdale (near R-65) (**Figure 10**), in water depths of approximately 40 ft (12 to 13 m) to serve as a receiver site for corals of opportunity.

4.2.2 Post-Construction As-Built Surveys of Coral Nursery Boulder Reefs

For each boulder reef, an as-built survey shall be conducted immediately following construction (within 6 months of construction) to determine the net acreage of reef constructed

4.2.2.1 As-Built Survey Goals

The main goal of an as-built survey is to estimate the net acreage of artificial hardbottom present within the recently constructed boulder reef footprint (gross delineated acreage). Gross and net acreage estimates along with all raw data used in calculations (*i.e.*, results of *in situ* hardbottom edge delineation and line-intercept surveys) shall be provided to the Department as part of the as-built survey effort for each boulder reef.

The estimated net acreage (see **Section 4.2.2.3** below) of the boulder reef shall be compared to the required mitigation acreage (set forth in the Permit and in **Section 4.2.3** based on UMAM analysis) to determine if

the required net acreage has been constructed. Net acreage shall be calculated as the product of the gross acreage (area within the footprint of the boulder reef) and the percentage of actual hardbottom within the boulder reef footprint. Gross acreage shall be measured by *in situ* delineation of the boulder reef boundary (see **Section 4.2.2.2** below).

4.2.2.2 *In Situ* Artificial Boulder Reef Boundary Delineation

The boundary of the boulder reef shall be delineated *in situ* as part of the as-built survey. For this task, divers shall swim the full extent of the edge of the boulder reef while towing a buoy equipped with a DGPS antenna attached by a cable to a Hypack navigation software system on a survey vessel. Once the boulder reef has been delineated, the acreage of the delineated area (= gross acreage) shall be determined.

4.2.2.3 Line-Intercept Survey Along Random Transects

The main goal of the line-intercept surveys is to determine the percentage of the area present within the boulder reef footprint (gross delineated acreage) that is accounted for by hardbottom. During the as-built survey, divers shall establish 10 m long temporary, randomly-placed transects, at a ratio of six (6) transects per quarter (0.25) acre of boulder reef. Prior to the *in situ* survey, the position of each temporary transect shall be determined by randomly generating a start point and degree heading. Divers shall use these GPS coordinates to locate each transect when conducting the survey. Line-intercept data for hardbottom cover (boulder area) shall be collected along the temporary transects. For this task, a diver shall swim the length of each temporary transect, note the location along the transect tape, and measure the linear extent of hardbottom (boulders) and all gaps between boulders. Following data collection, the percentage of each transect line accounted for by hardbottom (boulders) shall be calculated. The percentage of linear area covered by hardbottom along each transect shall be averaged across all transects surveyed at the reef to determine the mean percentage of hardbottom. This mean value shall be multiplied by the gross acreage (*in situ* delineated acreage) to arrive at an estimate of net hardbottom acreage for the boulder reef.

4.2.3 Boulder Reef Scleractinian Coral Nursery Mitigation Success Criteria

The following success criteria shall be met:

- C. 0.20 net acres of limestone boulder reef shall be constructed in the nearshore waters of the project area (near R-85) just south of the rubble barrier. This net acreage shall be documented during the as-built survey conducted immediately following (within 6 months) construction of the mitigation reef.
- D. 0.30 net acres of limestone boulder reef shall be constructed offshore of Fort Lauderdale (near R-65), in water depths of approximately 40 ft (12 to 13 m). This net acreage shall be documented during the as-built survey conducted immediately following (within 6 months) construction of the mitigation reef.

If after each as-built survey, mitigation criterion C or D, respectively, has/have not been met, then the

Permittee and/or authorized agent shall meet with Department staff to discuss a path forward to complete the mitigation requirement. Construction of additional mitigation acreage may be required to fulfill the mitigation requirement.

4.3 Relocating Corals and Actively Managing Coral Nursery Boulder Reefs Mitigation

Annual collection and relocation of “corals of opportunity” to the mitigation boulder reefs shall be conducted as a component of the coral nursery program (**Section 4.3.1**). Corals of opportunity are scleractinian corals that have been detached from the reef through natural processes or due to injury events for which there is no responsible party or corals on unconsolidated rubble on the portion of the spoil shoal seaward of the rubble removal area. Other sources of corals for the mitigation reefs include authorized maintenance dredging project footprints which do not require mitigation and permitted impact areas for which mitigation has already been constructed.

Relocation shall occur and be completed within 10 years of construction of the first artificial reef to be constructed, by which point a total of 1,500 coral colonies shall have been successfully established at the nursery boulder reefs. A minimum of 3,000 corals of opportunity shall be relocated to the mitigation boulder reefs as part of this effort. A subset of the coral colonies relocated to the artificial reef nurseries shall be monitored to determine the success of relocation effort (**Section 4.3.2**). The newly created mitigation boulder reef coral nurseries shall also be actively managed to promote the recruitment of corals and other benthic marine invertebrates (**Section 4.3.3**). Active management activities entail the cleaning (scrubbing) of boulder surfaces and the seeding of nurseries with sea urchins, when available, to increase the availability of bare substratum. The recruitment of scleractinian colonies to the mitigation reefs shall also be documented.

4.3.1 Collecting, Transporting, and Transplanting Corals of Opportunity

4.3.1.1 Collecting Corals of Opportunity

Corals of opportunity will be collected and relocated annually, but the specific methods and timing of coral collection will vary depending on the source of the corals of opportunity. If corals are relocated as an avoidance/minimization measure for coastal construction projects such as the planned Port Everglades Navigation Improvement Project, from berths projects within Ports (*e.g.*, Port of Palm Beach), or by way of corals less than 10 cm in diameter that are located landward of the permitted ETOF in beach nourishment projects in Broward County, then collection and relocation shall be coordinated carefully with project managers and regulatory agencies. All corals of opportunity relocated to mitigation boulder reef sites in a given year shall be treated as a cohort and shall be transplanted (**Section 4.3.1.3**) and monitored (**Section 4.3.2**) as such.

4.3.1.2 Transporting Collected Corals of Opportunity

If relocating corals of opportunity requires them to be transported (*i.e.*, colonies are not collected in close proximity to the mitigation receiver reef), then coral colony transportation shall be conducted as follows.

Care shall be taken to prevent injury to colonies while transporting them to their respective transplantation sites. Prior to transport, each colony shall be wrapped in bubble wrap packing material to prevent abrasion, and then stowed in a cooler on board the vessel to keep them cool, damp, and out of the sun. Colonies shall only be kept in coolers (out of the water) for a few hours. Colonies shall be transported to their respective transplantation sites each field day following collection. The aim shall be to transplant all colonies the same day they were collected. If conditions (*e.g.*, weather and sea state) do not allow for same day transplantation, then colonies will be transported to the transplantation site and cached in a safe, secure location. Attachment of the cached colonies shall be the priority for the next available field day.

4.3.1.3 Transplanting Collected Corals of Opportunity

Corals of opportunity shall be transplanted to boulders at their respective mitigation reef nursery sites. Each boulder to which colonies are transplanted shall be tagged so that cohorts of transplants can be identified. Individual boulders shall not contain mixtures of cohorts (*i.e.*, all corals on an individual boulder shall be of the same cohort). Colonies will be attached to boulders using Portland cement. Before attachment, both the boulder and coral (underside of the colony or rubble) will be prepared. Fouling organisms (mostly algae) and sediment will be removed from contact surfaces so that the cement can form a solid bond between the colony and the boulder. Wire brushes and chisels will be used to prepare surfaces. Care will be taken to avoid injury to living coral tissue during surface preparation. Corals already in place on the boulder reef shall be avoided by transplanting the relocated corals to the “upper/top” surface of boulders.

4.3.2 Monitoring Relocated Corals of Opportunity

To document the success of the transplantation process, the survival, health, and reproductive status of a subset of coral colonies in each cohort transplanted to each coral nursery boulder reef shall be tracked over time. All coral colonies transplanted to a given reef within a given year shall be treated as a cohort, and monitoring metrics shall be evaluated on a cohort by cohort basis. Each cohort shall be surveyed immediately following transplantation (baseline, $t=0$) and then again at years 1 and 3 post-transplantation. Individual colonies will not be tagged; instead, individual boulders shall be tagged. For monitoring the survival and health of colonies, the number of tagged boulders included in the monitoring effort for each cohort shall be sufficient to include at least 25% of colonies within each cohort. All colonies on boulders included in the monitoring effort shall be mapped and surveyed as specified in **Section 4.3.2.1**. For monitoring the reproductive status of colonies, the number of tagged boulders included in the monitoring effort for each cohort shall be sufficient to include at least 10% of the colonies within each cohort (at each time point) (see **Section 4.3.2.2**). The same boulders (*i.e.*, same colonies within a cohort) may be tracked when monitoring survival and health and also reproductive status.

4.3.2.1 Monitoring the Survival and Health of Relocated Corals of Opportunity

To document colony survival and health, 25% of the colonies in each cohort (number of colonies transplanted to the nursery each year) shall be surveyed immediately following transplantation (baseline) and then again at years 1 and 3 post-transplantation. During each survey of each cohort, colonies shall be identified to species, measured (maximum diameter), and have their condition recorded. The following shall

be noted for colony condition: percent live tissue; percent old and recent mortality; percent bleaching; and presence of disease, obvious signs of predation, and sedimentation (sediment accumulation). Thus, colony size and colony condition constitute the metrics for which data shall be collected during each survey. Whether a colony was relocated from rubble shall be noted during the baseline survey.

Raw data and reports shall be submitted to the Department after each annual survey. Reports shall include the presentation of metrics for the status of colonies within each cohort at each reef site and shall describe any changes in status from previous survey events. Survival, growth, and health data shall be presented in tabular and graphical formats. Survivorship, the number of live, attached colonies within a cohort at each time point (survey event), shall be reported for each cohort (and by species within each cohort). For each survey event, survivorship shall be expressed as the percent of surviving colonies within a cohort out of the total number of colonies in the cohort (number initially transplanted). Mean colony size, percent live tissue cover, percent old and recent mortality, and percent bleaching shall also be presented each survey for each cohort. Note that as colonies are lost to attrition or adverse conditions, mean values will need to be corrected for colonies that are lost (*i.e.*, the mean for each metric shall be calculated based on the remaining, surviving colonies at each survey). Presence of disease, obvious signs of predation, and sedimentation (sediment accumulation) shall also be described for each cohort for each survey and any changes in these parameters shall be discussed.

4.3.2.2 Monitoring the Reproductive Status of Relocated Corals of Opportunity

To document the reproductive status of nursery corals, 10% of the colonies in each cohort (number of colonies transplanted to the nursery each year) shall be evaluated through histological analysis at the time of collection (baseline) and at Year 3 post-transplantation. The colonies sampled at baseline (transplantation) need not be the colonies sampled at Year 3 post-transplantation, but the same species in the same abundances shall be sampled during the Year 3 post-transplantation survey. Each colony sampled in each cohort, at each time point, shall be identified to species, measured (diameter), and have its condition recorded. The following shall be noted for colony condition: percent live tissue; percent old and recent mortality; percent bleaching; and presence of disease, obvious signs of predation, and sedimentation (sediment accumulation).

Tissue samples (collected from 10% of colonies in each cohort at baseline and also at Year 3 post-transplantation) shall be preserved in Z-fix in the field for histological analysis. Samples shall be decalcified using a 10% hydrochloric acid solution buffered with EDTA. Decalcified tissue samples shall be processed, trimmed, and imbedded in paraffin blocks. Blocks shall be sectioned to prepare slides, and then slides stained. Using light microscopy, the slides shall be examined for the presence of gametes and gamete stages (Szmant-Froelich et al 1980). Colonies shall be considered reproductively active, and to be contributing to the surrounding population through larval supply, if any portion of the colony shows late stage gamete production (Stages III or IV) in histological samples. The percent of reproductively active colonies in each cohort at baseline (transplantation) and Year 3 post-transplantation shall be included in annual reports on the status of transplanted corals of opportunity.

4.3.3 Active Management of Coral Nursery Boulder Reefs

Coral nurseries constructed as mitigation shall be actively managed to promote colonization (sexual recruitment of corals and other benthic invertebrates) and coral colony transplantation success. For each boulder reef nursery site, the following tasks shall be performed each year prior to coral spawning events that generally occur in August/September: cleaning reef surfaces to promote larval settlement (**Section 4.3.3.1**) and seeding nurseries with sea urchins as a natural source of continuous algae control (**Section 4.3.3.3**). The settlement of coral larvae on the boulder reefs will also be tracked over time (**Section 4.3.3.2**).

4.3.3.1 Cleaning the Surfaces of Nursery Boulder Reefs and Tracking Coral Recruitment

To reduce the overgrowth of turf and macroalgae on reef boulders, the surfaces of boulders at each mitigation boulder reef coral nursery shall be cleaned once per year for five (5) consecutive years (N=5 cleaning events per boulder reef). During each cleaning event, wire brushes shall be used to remove turf and macroalgae from 25 cm diameter areas (circles) around each transplanted coral colony at each boulder reef. Cleaning shall start one (1) year after construction and end six (6) years after the construction (Years 1 - 6 post-construction for each reef) of each reef, respectively. Each cleaning event shall be performed two weeks prior to the annual coral spawning event in August/September.

4.3.3.2 Tracking the Recruitment of Scleractinian Corals to the Nursery Boulder Reefs

Recruitment (sexual) of scleractinian corals to the mitigation reef shall be assessed during coral survival/growth/health surveys (see **Section 4.3.2.1**) and shall continue for as long as survival/growth/health surveys of cohorts of transplanted colonies are conducted. The area within 25 cm of all relocated colonies on tagged boulders shall be searched and all coral recruits (non-transplanted colonies greater than 2 cm in maximum dimension) shall be recorded and identified to the lowest taxonomic level possible (to species, if possible). Recruitment data shall be included in monitoring reports submitted to the Department.

4.3.3.3 Seeding Coral Nursery Boulder Reefs with Sea Urchins

As part of active management of the coral nursery, mitigation boulder reefs shall be seeded with long-spined sea urchins (*Diadema antillarum*) for three consecutive years after construction (N=3 relocation events per boulder reef). Urchin relocation each post-construction year is dependent on the availability of urchins (*i.e.*, presence and abundance of *Diadema antillarum*) in the collection area and relocation shall not be required any year where fewer than 5 urchins (per artificial reef) are available. Urchins shall be collected from the rubble habitats at the spoil shoal, and though the current abundance of urchins in this area is unknown, juvenile *Diadema antillarum* were observed in rubble habitats at the spoil shoal during surveys in 2007 and 2013.

For each mitigation boulder reef coral nursery, the relocation of long-spined sea urchins shall start one (1) year after construction and end three (3) years after construction (Years 1 - 4 post-construction for each reef). For three years following the construction of each artificial reef, respectively, monitoring team staff shall search rubble habitats at the spoil shoal for *Diadema antillarum* during annual collection of corals of

opportunity. If at least five (5) urchins are present (per artificial reef), then all urchins located during the search shall be relocated to the mitigation boulder reefs. Relocation shall occur in summer months prior to annual coral spawning events.

4.3.4 Relocated Corals of Opportunity and Active Nursery Management Mitigation Success Criteria

The following success criteria shall be met:

- E. A minimum of 3,000 scleractinian corals of opportunity shall be transplanted to the boulder reef coral nursery mitigation sites by Year 10 post-construction of the first mitigation boulder reef to be constructed.
- F. A total of 1,500 scleractinian corals of opportunity shall be successfully established at the boulder reef coral nursery mitigation sites by Year 10 post-construction of the first mitigation boulder reef to be constructed. To meet this criterion, the coral colony survival rate (survivorship) shall be at least 50% within each cohort at each mitigative reef by each cohort's final (Year 3 post-transplantation) monitoring event or the weighted mean coral colony survival rate (mean weighted by the number of colonies in each cohort at each mitigative artificial reef) shall be at least 50% after completion of the final (Year 3 post-transplantation) monitoring event for the last cohort relocated to the boulder reef coral nursery mitigation sites. To be counted as surviving, a colony shall be (have remained) securely attached to the substratum at the final (Year 3 post-transplantation) monitoring event.
- G. For each cohort of relocated corals at each mitigative reef, mean percent live tissue cover of the surviving coral colonies shall be at least 30% of that measured at baseline (transplantation) three (3) years after transplantation (i.e., at the Year 3 post-transplantation monitoring event).
- H. For each cohort of relocated corals, the percentage of reproductively active colonies with late stage gametes three (3) years after transplantation (year 3 post-transplantation for each cohort at each mitigative reef) shall be at least 10% of the percentage documented at the time of collection (baseline).
- I. Starting the year after construction (for each boulder reef, respectively), the surfaces of boulders at each mitigation boulder reef coral nursery shall be cleaned with wire brushes once/annually within two weeks prior to the annual spawning event for five (5) consecutive years to remove turf and macroalgae.
- J. Starting the year after construction (for each boulder reef, respectively), if there is a source of long-spined urchins within the rubble habitats at the spoil shoal, the mitigation boulder reef coral nurseries shall be seeded with long-spined sea urchins once per year for three (3) consecutive years. Long-spined urchins shall be collected from post-construction rubble habitats and relocated to nursery boulder reefs.

If by Year 10 post-transplantation, all mitigation success criteria set forth in **Section 4.3.4** (criteria E – J) have not been met, then the Permittee and/or authorized agent shall meet with Department staff to discuss the performance of the mitigation and a path forward to complete mitigation requirements. Additional mitigation (*e.g.*, boulder reef construction, coral colony transplantation, reef scrubbing, etc.) may be required to fulfill mitigation requirements. Events that impact the local coral community (*e.g.*, regional bleaching, disease outbreaks, and hurricanes) shall be considered when evaluating the success of relocated coral colonies. Southeast Florida Coral Reef Evaluation and Monitoring Project (SECREMP) inner reef tract and middle reef tract survey sites (N=2 sites with N=4 stations per site) in Broward County, FL (SECREMP Sites BC1 [26° 08.872' N, 80° 05.758' W] and BC2 [26° 09.597' N, 80° 04.950' W]) shall serve as local reference sites for coral health. Inner reef tract and middle reef tract Broward County sites surveyed by the Florida Reef Resilience Program (FRRP) Disturbance Response Monitoring program may also be used as additional reference sites for coral health. The Permittee is responsible for providing the Department with annual data collected by these programs in the areas identified.

5.0 BEACH FILL PLACEMENT MONITORING

The FDEP permit for the Port Everglades Sand Bypassing project provides for placement of beach-compatible material dredged from the Port Everglades sand trap in the template at Dr. Von D. Mizell-Eula Johnson State Park (between R-86 and R-92) (**Figures 1-4**). The permitted ETOF for the Port Everglades Sand Bypassing project is the same as that permitted for the 2003 Broward County Segment III project (FDEP Permit No. 0163435-001-JC, Expired 5/12/2008). Biological monitoring for the first beach fill placement event conducted under the Port Everglades Sand Bypassing project permits (FDEP Permit No. 0289308-004-JC and USACE Permit No. SAJ-2008-02034 [SP-LCK]) shall include a pre-construction (baseline) survey during the summer prior to fill placement, an initial post-construction survey (within six months of fill placement), and three annual post-construction surveys (Years 1, 2, and 3 post-construction) following fill placement. Each subsequent fill placement event (nourishment) conducted under this permit shall initiate a complete round of post-construction monitoring (*i.e.*, an initial post-construction survey and three annual post-construction surveys at Years 1, 2, and 3). The pre-construction survey conducted prior to the first fill placement event shall serve as the baseline for all post-construction monitoring events, and no additional pre-construction surveys shall be required prior to subsequent fill placement events. Standard operating procedures shall be used during each monitoring event to provide consistent and repeatable monitoring to document sediment dynamics and changes in hardbottom communities.

The aim of these surveys is to document any unpermitted direct and/or secondary adverse impacts to nearshore hardbottom resources that may occur due to the spreading of project sand outside of the permitted ETOF (*i.e.*, further than predicted). As such, surveys will document sediment depth and cover as well as the abundance, distribution, condition, and function of hardbottom resources (biotic assemblages) beyond (outside of) the permitted ETOF during each monitoring event. Nearshore hardbottom adjacent to the fill template, but outside of the permitted ETOF (between R-86 and R-92) for this project, shall be monitored (**Figures 1-4**). Hardbottom monitoring shall consist of nearshore hardbottom edge mapping and transect monitoring as described below.

5.1 Nearshore Hardbottom Edge Mapping and Monitoring

Mapping of the nearshore hardbottom edge by divers will provide information on hardbottom exposure within the project area and allow for determination of direct impacts due to hardbottom burial by project fill (should impacts occur). The entire length of the nearshore hardbottom edge between R-86 and R-92 shall be mapped during the pre-construction (baseline) monitoring event and in all post-construction monitoring events. Continuous, annual delineation of the hardbottom edge is also permissible. The nearshore hardbottom edge is defined as the visible border between sand and hardbottom. Bounce dives shall not be used to delineate the hardbottom edge unless authorization is granted by FDEP staff. If necessary, requests shall be submitted to the JCP Compliance Officer (JCPCompliance@dep.state.fl).

The method used for *in situ* mapping shall consist of divers swimming the entire length of the hardbottom edge, with one diver towing a DGPS antenna transmitting continuous positions to HYPACK hydrographic survey or GIS software on board a survey vessel. The towing-diver shall swim at a speed conducive for maintaining the buoy on as short a tether as possible above him/her.

5.2 Establishment and Monitoring of Permanent Transects

Hardbottom monitoring adjacent to the fill template (outside/seaward of the permitted ETOF) shall be conducted using permanent, shore-perpendicular transects of two types — biological and sediment-only (**Figure 18**). All surveys (line-intercept, interval sediment depth measurements, benthic quadrat surveys, and qualitative video surveys [Sections 5.2.2 – 5.2.5]) apply to biological transects, while only line-intercept and interval sediment depth measurements (Sections 5.2.2 and 5.2.3) apply to sediment-only transects. To obtain the most accurate sediment depth and location information, line-intercept surveys and interval sediment depth measurements shall be conducted first along each transect during each monitoring event.

5.2.1 Establishment of Permanent Transects and Quadrats

A total of 12 permanent, 150-m, shore-perpendicular transects between R-86 and R-92 shall be used to monitor nearshore hardbottom resources adjacent to the fill template for the Port Everglades Sand Bypassing Project (**Figure 18**). Six (6) transects (BT1-BT6) shall be full biological transects and six (6) transects (ST1 – ST6) shall be sediment-only transects (**Table 2 and Figure 18**). Monitoring of resources within the permitted ETOF is not a component of permit-required biological monitoring; as such, each transect shall start at the nearshore hardbottom edge. If the permitted ETOF is located seaward of the nearshore hardbottom edge (*i.e.*, the hardbottom edge is within the ETOF), then transects shall start at the permitted ETOF. Thus, the entire length of each transect shall be outside the permitted ETOF. Once established, the 12 transects shall be permanent, and their positions and lengths shall not change over time. Permanent markers (pins, iron rods, etc.) shall be installed at the start, end, and every 25 m to 50 m along each transect line to ensure repeatability in transect placement during each monitoring event. Permanent markers shall be maintained over time (*e.g.*, replaced when worn out or lost) to further aid in correct transect placement.

A total of 15 permanent quadrats measuring 1 m² each shall be established along each biological transect (15 m² per transect). The first permanent quadrat along each biological transect shall be installed at meter zero (0) (nearshore hardbottom edge or ETOF). From this point, quadrats shall be established at roughly the 5, 10, 15, 20, 25, 30, 40, 50, 60, 75, 90, 110, 130, and 150-meter marks along each transect. If the distribution of sand along a transect line interferes with the placement of a quadrat in one of the specified locations, then the quadrat shall be installed in hardbottom as close as possible to the specified location (*i.e.*, the positions of all quadrats will be initially established such that areas covered by sand are avoided). Once installed during the pre-construction (baseline) survey, the positions of all quadrats shall be permanent, and post-construction surveys shall use the same quadrat locations as the pre-construction survey, regardless of the exposure or burial condition of each location after initial quadrat establishment. To ensure the same quadrats in the same positions are surveyed during each monitoring event, the northeast corner of each quadrat will align with a particular meter mark on each transect, and pins (or nails or eye-bolts) will be installed to permanently mark the location of each quadrat. The permanent location of each quadrat shall be recorded and reported for each survey, and post-construction survey quadrat positions shall match baseline quadrat positions.

Table 2. Transect information including name, nearest R-monument, type, and length.

Transect Name	Nearest R-Monument	Transect Type	Transect Length (m)
BT-1	R-86.5	Biological	150
ST-1	R-87	Sediment Only	150
BT-2	R-87.5	Biological	150
ST-2	R-88	Sediment Only	150
BT-3	R-88.5	Biological	150
ST-3	R-88.5	Sediment Only	150
BT-4	R-89	Biological	150
ST-4	R-89.5	Sediment Only	150
BT-5	R-90	Biological	150
ST-5	R-90.5	Sediment Only	150
BT-6	R-91	Biological	150
ST-6	R-92	Sediment Only	150

5.2.2 Line-Intercept Survey

To document larger areas of uninterrupted sand (physical transitions along the monitoring transects between sand vs. hardbottom), and to track changes in sediment cover over the hardbottom, line-intercept surveys shall be conducted along each permanent transect. During each monitoring event, the landward and seaward position of each sand patch/trough larger than 0.5 m in length shall be recorded along each transect by reference to transect tape meter marks. Meter mark references shall be to one decimal place (*e.g.*, patch from 2.4 to 3.2 m).

5.2.3 Interval Sediment Depth Measurements

To track changes in sediment depth associated with changes in sediment cover, each monitoring event shall include collection of interval sediment depth measurements along each permanent transect. Sediment depth

shall be measured at 1-m intervals along the entire length of each transect, inclusive of sand patches. For each measurement, a stainless-steel ruler graduated in centimeters (0 to 30 cm) shall be pressed through the sediment until the ruler reaches the surface of hard substrata or is totally immersed in sand. Depth measurements shall be rounded to the nearest cm (*i.e.*, sediment thickness of less than 0.5 cm will be recorded as “0 cm”, while thickness greater than 0.5 cm but equal or less than 1 cm will be recorded as “1 cm”, etc.). Measurements greater than 30 cm shall be recorded as “> 30 cm”.

5.2.4 Quadrat Sampling

Benthic communities and their habitats shall be characterized quantitatively using the quadrat method, which includes sampling habitat and assemblages within permanently positioned 1 m² quadrats along each biological transect. This method ensures that the same quadrats (same location, same size) are sampled during each monitoring event to document changes in sediment and communities over time. The sampling protocol is similar to that used in the Benthic Ecological Assessment for Marginal Reefs (BEAMR) (Lybolt and Baron, 2006). As described below, three main benthic characteristics shall be assessed in each quadrat during sampling: physical structure, planar percent cover of sessile benthos, and coral (scleractinian and octocoral) size and density.

For physical structure, maximum topographic relief and mean sediment depth (average of three depth measurements) shall be measured in centimeters (cm) within each quadrat. For functional groups, planar percent cover estimates of all sessile benthos and substratum features shall be made within each quadrat; the following 21 functional groups shall be assessed: sediment (*e.g.* sand, shell-hash, or mud), bare hardbottom, rubble, macroalgae, encrusting red algae, turf algae, cyanobacteria, sponge, octocoral, scleractinian coral, anemone, zoanthid, hydroid, hydrocorals (*e.g.*, *Millepora* spp.), sessile annelid (not including *Phragmatopoma* spp.), wormrock, barnacle, bivalve, bryozoan, echinoderm, and tunicate. Each functional group will be assigned a percent cover value from 0 to 100%, with the total of all functional groups in each quadrat equaling 100%. Only cyanobacteria covering hardbottom or benthic organisms shall be included in quadrat percent cover estimates. Cyanobacteria cover over sand (most often as mats) will be noted and assessed, but these data will not to be included in the quadrat percent cover estimates, as doing so could obscure the percent cover value of the sediment category. Macroalgae shall be identified to genus level and the percent cover of each genus with cover greater than 1% shall be recorded; unattached or floating macroalgae shall be disregarded. Scleractinian coral colonies shall be identified to species and octocoral colonies will be identified to genus. The identities of known common sponge species/genera shall be listed (*e.g.*, *Pione lampa*, *Cliona deletrix*, *C. varians*, *Cinachyrella apion*, *Desmapsamma anchorata*, etc.).

Monitoring staff shall also measure and record to the nearest centimeter (cm) the maximum dimension (height or width) of each scleractinian coral and octocoral colony within each quadrat. The smallest size recorded shall be one (1) cm; for colonies less than one (1) cm in size, the measurement recorded will be “< 1 cm”. Each colony within each quadrat shall also be enumerated and identified (by species for scleractinians, by genus for octocorals) to determine coral density and composition.

5.2.5 Qualitative Video Survey

Qualitative video surveys serve as archival data sets for reference or further resolution of potential impacts suggested by quadrat and sediment survey data. As such, survey data could be reviewed and compared between surveys to document qualitative changes along transects over time for refining the impact area assessment. Qualitative video surveys shall be conducted along all 12 monitoring transects using a high definition digital video camera in a waterproof housing. Video of the seafloor along each transect shall progress no faster than 5 meters per minute (3.2 inches/second or about ½ hr. per 150 m transect). During the survey, a convergent laser guidance system shall be used to precisely maintain the height of the camera at 40 cm above the bottom. The transect line shall be clearly visible in all video so that locations may be accurately referenced. A 360° panoramic view at an angle of roughly 30° to the horizon shall be recorded both at the beginning and end of each transect from an elevation of roughly 1 m above the bottom. At the beginning and end of each transect, a standard underwater display shall also be recorded and integrated directly onto the digital video track. The standard display shall report: 1) the project permit number (standard DEP/JCP project number); 2) the transect name/number; 3) the survey date; 4) the water depth in meters for both the beginning (transect meter 0) and end (meter 150) of the transect; and 5) any pertinent notes (e.g., poor visibility, large swell, etc.). Video data (files) shall be supplied to FEDP during raw data submittal.

6.0 MONITORING STAFF AND QA/QC REQUIREMENTS AND SCHEDULE

The names and qualifications of the monitoring staff performing the monitoring shall be submitted by the Permittee or their Agent to the FDEP for approval. Written agency approval of personnel will be required prior to proceeding with the yearly monitoring. Biological monitoring surveys shall be conducted by staff with previous experience in monitoring hardbottom communities and with scientific knowledge of local benthic marine ecosystems and flora and fauna. All in-water crew members responsible for *in situ* quadrat data collection shall participate in cross training to verify correct species identification and survey practices as Quality Assurance/Quality Control (QA/QC) procedures at the beginning of each monitoring event. QA/QC results shall reflect consistency of 90% for percent cover and identification of functional groups between observers. All permit required monitoring data submitted to the Department shall have been checked against field datasheets and corrected (if necessary) to ensure accuracy.

See each of the four monitoring/minimization/mitigation sections (**Sections 2.0 - 5.0**) for task specific monitoring event timelines/schedules. Summary tables for activities and monitoring required in **Sections 2.0 – 5.0** are provided in **Appendices A – E**.

7.0 REPORTING

7.1 Notification of Commencement, Progress, and Completion of Work

Commencement dates of minimization/avoidance activities and monitoring events and surveys (work) shall be reported via email to the JCP Compliance Officer (JCPCompliance@dep.state.fl) roughly seven (7) days

prior to the start of in-water work, and the day that work begins. Brief work progress summaries shall be submitted (emailed) weekly to the JCP Compliance Officer until the completion of the JCP Compliance Officer shall also be notified when work has been completed.

7.2 Monitoring Data Submission

For each data submission event, data shall be provided in a standardized format, as specified below in **Section 7.2.4**.

7.2.1 Pre-Construction Data

7.2.1.1 Relocated Scleractinian and Octocoral (Minimization) Monitoring Data

At least 30 days prior to construction (initial excavation) of the sediment impoundment basin and within 60 days of completing coral relocation and baseline coral monitoring the monitoring firm shall submit all baseline data for scleractinian corals and octocorals relocated from the direct and indirect impact areas (see **Section 2.0**) directly and concurrently (simultaneously) to the FDEP JCP Compliance Officer, the USACE, and the Permittee/Agent in electronic format. Data shall be submitted either on a portable hard drive or via an FTP site.

7.2.1.2 Sand Trap Excavation and Maintenance Dredging Monitoring Data

At least 30 days prior the commencement of any and all sand trap related construction activities (excavation or maintenance dredging) and within 60 days of completing pre-construction (excavation or maintenance dredging) sediment trap related monitoring (*e.g.*, baseline surveys of transects in and around the turbidity mixing zone [see **Section 3.0**]) the monitoring firm shall submit all raw pre-construction (baseline) data directly and concurrently (simultaneously) to the FDEP JCP Compliance Officer, the USACE, and the Permittee/Agent in electronic format. Data shall be submitted either on a single portable hard drive or via an FTP site.

7.2.1.3 Fill Placement Nearshore Hardbottom Monitoring Data

At least 30 days prior to the commencement of beach fill placement related construction activities and within 45 days of completing pre-construction (baseline) nearshore hardbottom monitoring (see **Section 5.0**) the monitoring firm shall submit all raw pre-construction (baseline) data directly and concurrently (simultaneously) to the FDEP JCP Compliance Officer, the USACE, and the Permittee/Agent in electronic format. Data shall be submitted either on a portable hard drive or via an FTP site.

7.2.2 During-Construction Data

7.2.2.1 Sand Trap Excavation and Maintenance Dredging Hardbottom Monitoring Data

Raw, during-construction (excavation or maintenance dredging) sedimentation monitoring data shall be

submitted by the monitoring firm directly and concurrently (simultaneously) to the FDEP JCP Compliance Officer, the USACE, and the Permittee/Agent in electronic format within 7 days of completing each required weekly survey (see **Section 3.0**). Data shall be submitted either on a portable hard drive or via an FTP site.

7.2.2.2 Sand Trap Excavation Seagrass Monitoring Data

If triggered during excavation of the sand trap, seagrass bed edge mapping and visual inspection data (see **Section 3.1.4**) shall be submitted directly and concurrently (simultaneously) to the FDEP JCP Compliance Officer, the USACE, and the Permittee/Agent in electronic format within 14 days of completing the required survey.

7.2.3 Post-Construction Data

7.2.3.1 Relocated Scleractinian and Octocoral (Minimization) Monitoring Data

Within 60 days of completing each post-transplantation monitoring event the monitoring firm shall submit all data for scleractinian corals and octocorals relocated from the direct and indirect impact areas (see **Section 2.0**) directly and concurrently (simultaneously) to the FDEP JCP Compliance Officer, the USACE, and the Permittee/Agent in electronic format. Data shall be submitted either on a portable hard drive or via an FTP site.

7.2.3.2 Sand Trap Monitoring Data

7.2.3.2.1 Sand Trap Excavation and Maintenance Dredging Hardbottom Monitoring Data

Within 60 days of completing sand trap related post-construction (excavation or maintenance dredging) sedimentation monitoring (see **Section 3.0**), the monitoring firm shall submit all raw data directly and concurrently (simultaneously) to the FDEP JCP Compliance Officer, the USACE, and the Permittee/Agent in electronic format.

7.2.3.2.2 Sand Trap Excavation Seagrass Monitoring Data

Within 45 days of completing sand trap excavation related post-construction (excavation or maintenance dredging) seagrass monitoring (see **Section 3.1.4**), the monitoring firm shall submit all raw data directly and concurrently (simultaneously) to the FDEP JCP Compliance Officer, the USACE, and the Permittee/Agent in electronic format.

7.2.3.3 Mitigation Monitoring Data

7.2.3.3.1 Re-exposure of Hardbottom Monitoring Data

Within 60 days of completing each post-construction monitoring event (see **Section 4.1**), the monitoring

firm shall submit all raw data associated with desktop aerial analysis, *in situ* delineation and ground-truthing (**Section 4.1.1.1**), and *in situ* benthic community monitoring (**Section 4.1.1.2**) directly and concurrently (simultaneously) to the FDEP JCP Compliance Officer, the USACE, and the Permittee/Agent in electronic format. Data shall be submitted either on a portable hard drive or via an FTP site.

7.2.3.3.2 Boulder Reef Scleractinian Coral Nursery Monitoring Data

Within 30 days of completing each post-construction as-built survey (see **Section 4.2.2**), the monitoring firm shall submit all raw data directly and concurrently (simultaneously) to the FDEP JCP Compliance Officer, the USACE, and the Permittee/Agent in electronic format. Data shall be submitted either on a portable hard drive or via an FTP site.

7.2.3.3.3 Transplanted Scleractinian Coral Monitoring Data

Within 60 days of completing each annual post-transplantation monitoring event (see **Sections 4.3.2**), the monitoring firm shall submit all raw data collected on coral the survival and health and on the reproductive status of colonies directly and concurrently (simultaneously) to the FDEP JCP Compliance Officer, the USACE, and the Permittee/Agent in electronic format. Data shall be submitted either on a portable hard drive or via an FTP site. Along with these data, the monitoring firm shall also provide raw coral recruitment data (**Section 4.3.3.2**).

7.2.3.4 Fill Placement Nearshore Hardbottom Monitoring Data

Within 45 days of completing each post-construction nearshore hardbottom monitoring event (see **Section 5.0**) the monitoring firm shall submit all raw data directly and concurrently (simultaneously) to the FDEP JCP Compliance Officer, the USACE, and the Permittee/Agent in electronic format. Data shall be submitted either on a portable hard drive or via an FTP site.

7.2.4 Raw Data Formats

Raw data provided to the FDEP shall consist of the following, each of which are described below: aerial photographs, video and photographs, hardbottom edge survey data; raw transect survey data, and field datasheets.

7.2.4.1 Aerial Photographs

Aerial photography shall include georeferenced, tiff format products along with the projected coordinate system and datum used to georeference each image. Aerial photographs will be collected annually, with the most recent set submitted along with that year's monitoring data (see **Sections 4.1.1.1 and 7.2.3.3.1**).

7.2.4.2 Video and Photographs

Quantitative and qualitative digital video and still photographs shall be submitted to the Department using

portable, large capacity, hard drives. When submitted together, video and still photographs shall appear in separate folders on the hard drive. Main folders will be identified by a descriptive name, so data may be easily differentiated. Each main folder will contain separate labeled subfolders when multiple data sets are submitted.

7.2.4.3 Hardbottom Edge and Other Boundary Delineation Survey Data

Edge and boundary mapping data shall be supplied as a collection of shapefiles (*e.g.*, as an ESRI file geodatabase). Lines or polygons shall represent the *in situ* mapped edge or patch from each survey. Depending whether an edge or a boundary is delineated, these data may be depicted as a single line (or set of lines) or as a polygon (or polygons). The permitted ETOF at Dr. Von D. Mizell-Eula Johnson State Park shall be submitted along with project deliverables following each monitoring event.

A separate collection of shapefiles (*e.g.*, as an ESRI file geodatabase) shall be provided for different datasets. Similar attributes as those described above shall be provided with these data.

7.2.4.4 Transect Survey Data

Interval sediment depth measurements, line-intercept data, quantitative photo survey data, BEAMR quadrat data, and belt transect data collected during monitoring events shall be supplied in Excel format. Separate Excel workbooks shall be supplied for multiple, disparate data sets. Each workbook submitted shall have a descriptive name, so data can be easily differentiated.

7.2.4.5 Field Datasheets and Survey Logs

Copies (photographs or scans) of field datasheets shall be submitted in pdf format.

7.3 Monitoring Report Submissions

Monitoring reports shall be provided for all permit required monitoring. Along with each monitoring report, the data (*e.g.*, collected pre-, during-, and post-construction) analyzed to produce the report shall be submitted along with tables used in the analysis of data, tables used to construct figures, and tables and figures provided in the report (in Excel format). Tables entered into Primer and Primer analysis files shall also be submitted. Each monitoring report shall clearly describe methods used in monitoring and data analysis and explain any deviations from the monitoring plan or conditions of the permit. Reports shall also provide results in appropriate graphical, tabular, and text formats.

7.3.1 Relocated Scleractinian and Octocoral (Minimization) Monitoring Reports

Within 90 days of completing each post-transplantation monitoring event, the monitoring firm shall submit a written report on the health and status of scleractinian corals and octocorals relocated from the direct and indirect impact areas (see **Section 2.0**) directly and concurrently (simultaneously) to the FDEP JCP Compliance Officer, the USACE, and the Permittee/Agent in electronic format.

7.3.2 Sand Trap Turbidity Mixing Zone Area Related Monitoring Reports

Within 90 days of completing post-construction monitoring for the initial excavation of the sand trap and within 90 days of completing post-construction monitoring for each maintenance dredging event, a written report on the results of hardbottom (and seagrass) sedimentation monitoring (see **Section 3.0**) directly and concurrently (simultaneously) to the FDEP JCP Compliance Officer, the USACE, and the Permittee/Agent in electronic format.

7.3.2 Mitigation Monitoring Reports

Within 90 days of completing yearly required post-re-exposure, post-construction, and post-transplantation surveys, respectively (see **Section 4.0**), written reports shall be supplied directly and concurrently (simultaneously) to the FDEP JCP Compliance Officer, the USACE, and the Permittee/Agent in electronic format. Each report shall indicate and describe the degree to which each aspect of mitigation has succeeded in reaching its success criteria.

7.3.3 Nearshore Hardbottom Beach Fill Placement Monitoring Reports

Within 90 days of completing each post-construction monitoring event, the monitoring firm shall submit a written report on the results of beach fill placement nearshore hardbottom monitoring (see **Section 5.0**) directly and concurrently (simultaneously) to the FDEP JCP Compliance Officer, the USACE, and the Permittee/Agent in electronic format. Note that no report is required to be submitted following the pre-construction (baseline) survey (*i.e.*, only raw data are required to be submitted to the FDEP).

8.0 FIGURES

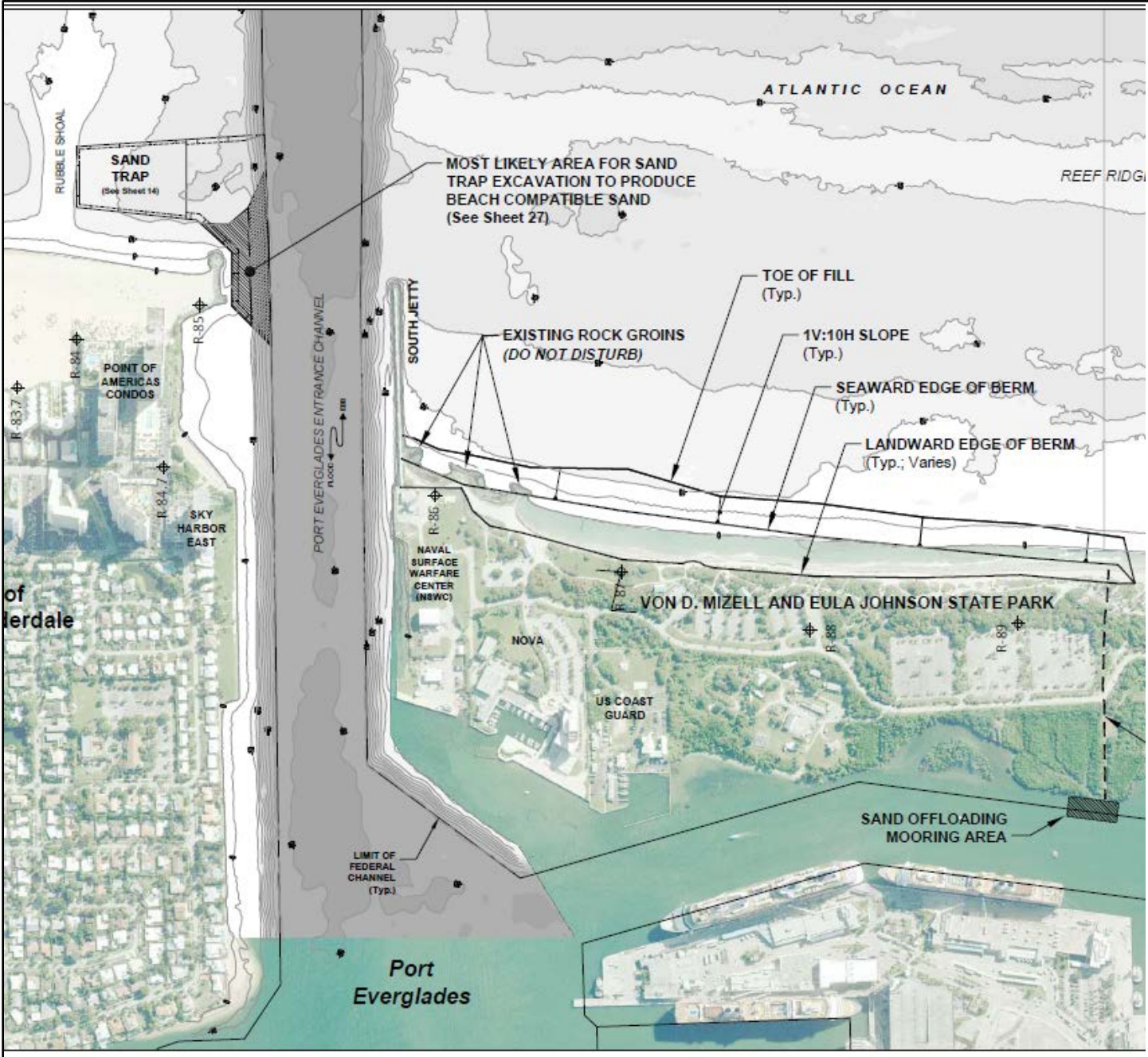


Figure 1. Port Everglades Entrance Channel and surrounding areas.

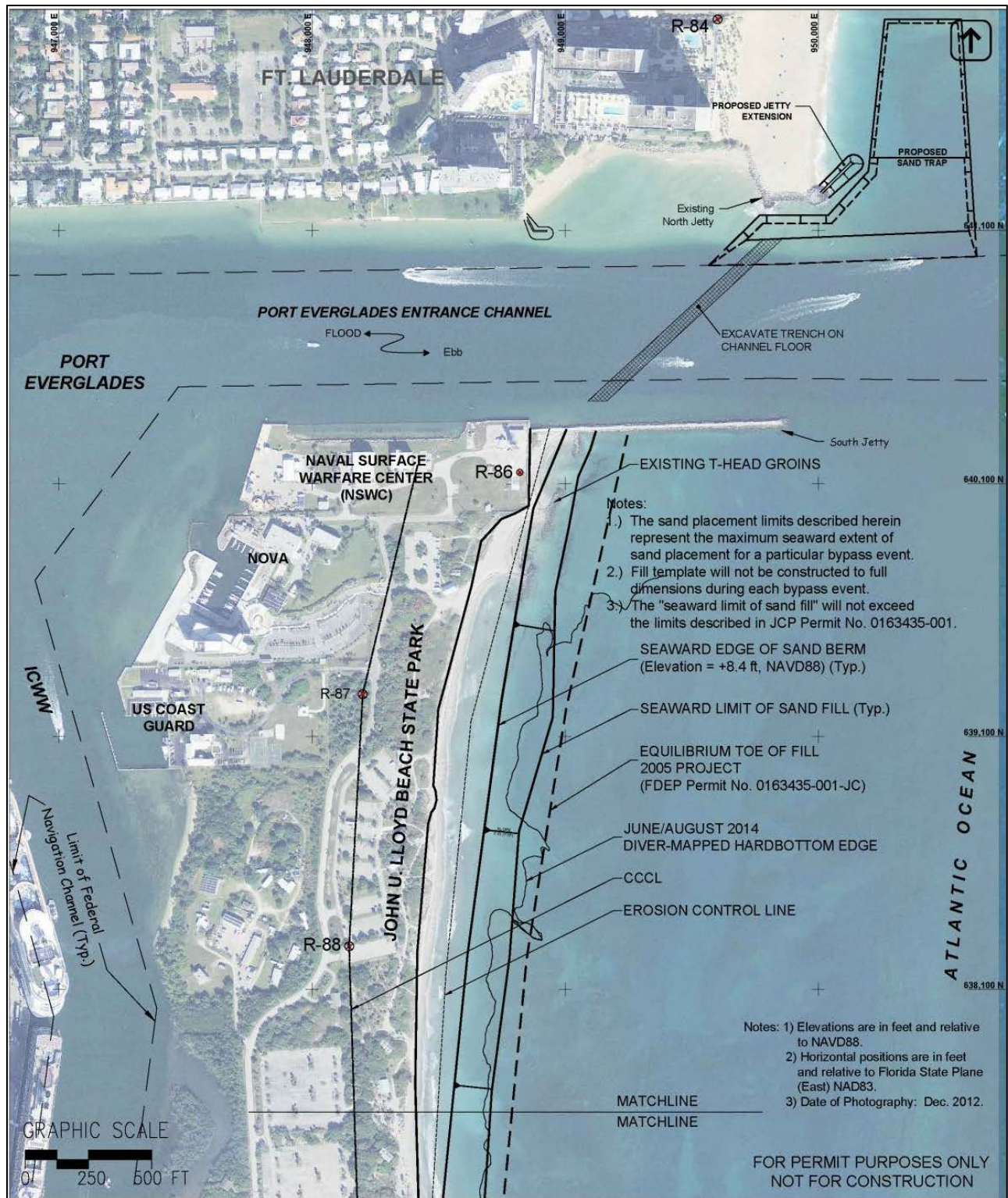


Figure 2. Location of the sediment impoundment basin (channel side north) and the sand placement area (south of channel) within the northern section of the Dr. Von D. Mizell-Eula Johnson Beach State Park (Source: OAI, 2015).

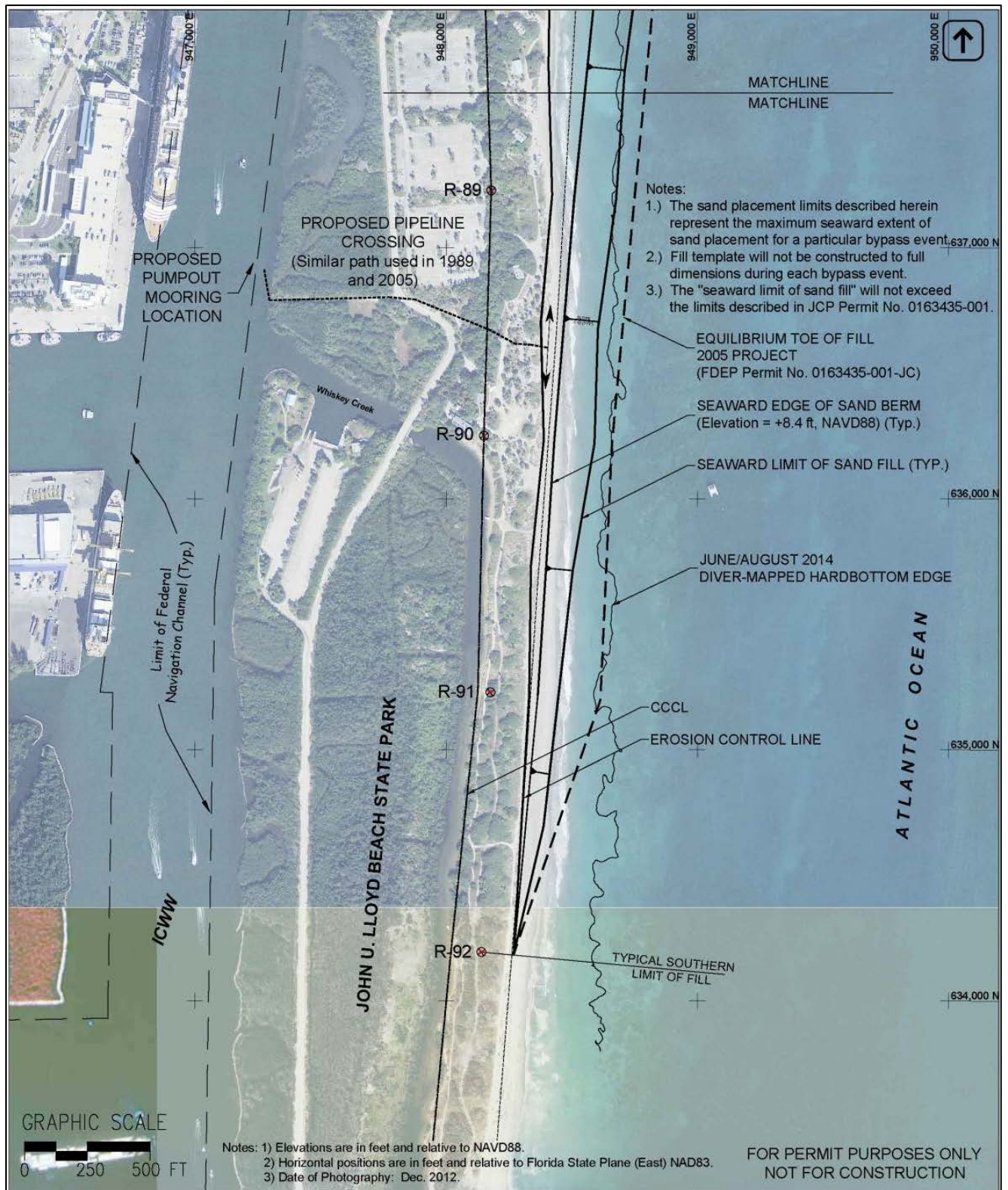


Figure 3. Sand placement area within the southern section of the Dr. Von D. Mizell-Eula Johnson Beach State Park (formerly the John U. Lloyd State Park) (Source: OAI, 2015).

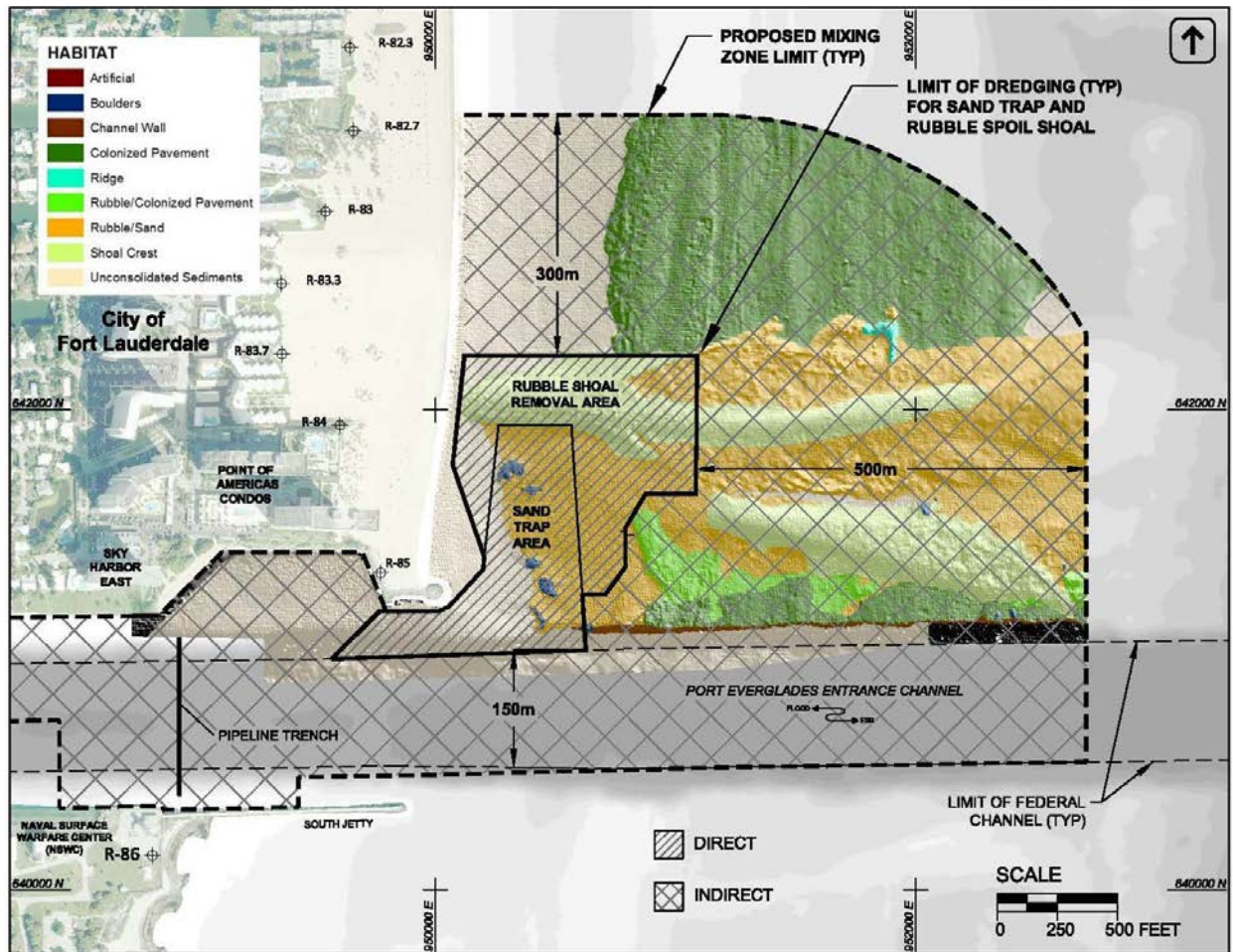


Figure 4. Port Everglades Inlet Channel project area benthic habitat map with direct impact areas (10.8-acre sediment impoundment basin and western portion of the rubble spoil shoal removal area) and indirect impact areas (areas within the turbidity mixing zone) outlined. Inset legend identifies habitat types.

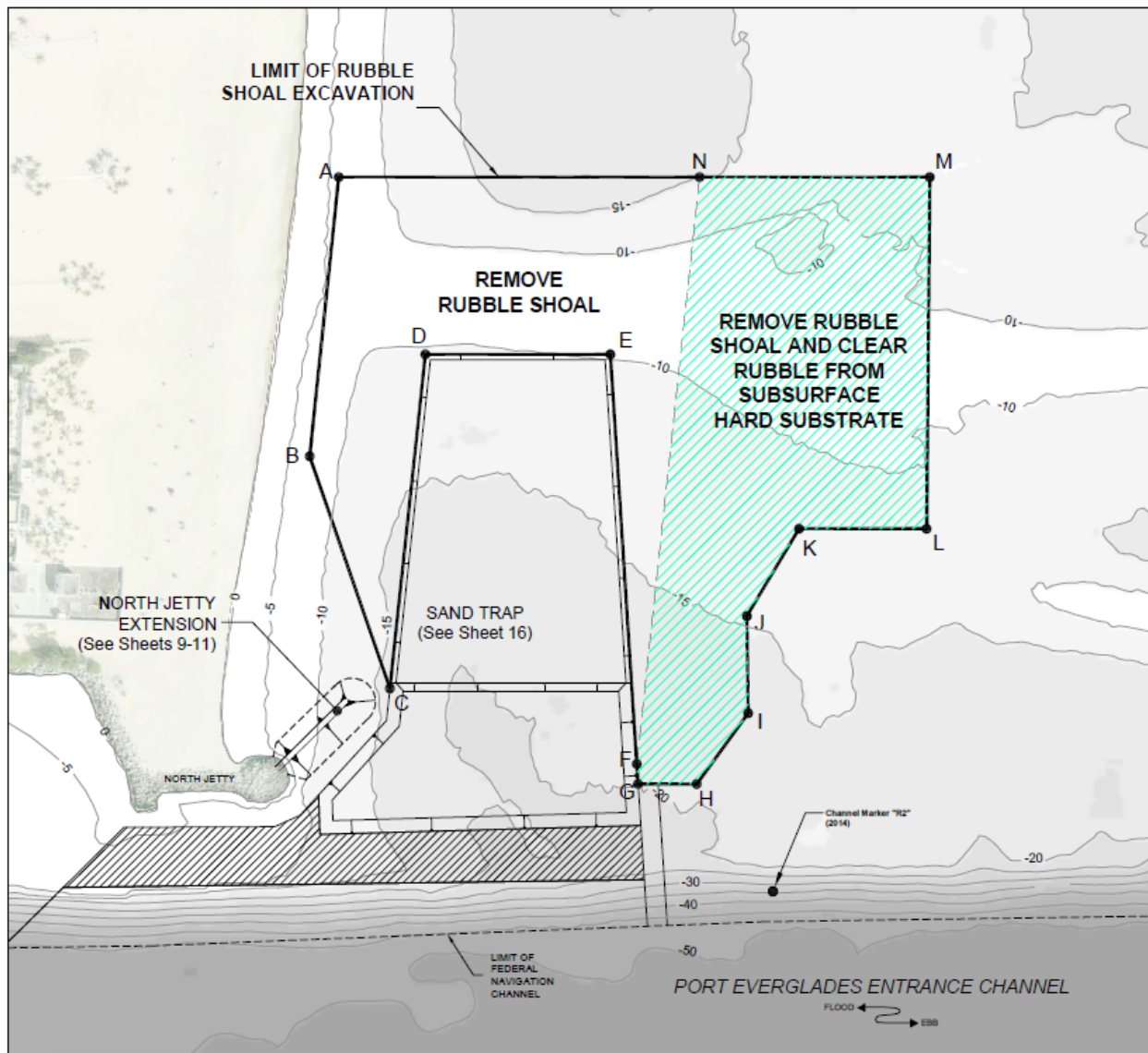


Figure 5. North jetty extension (160 feet towards the northeast) at R-85.

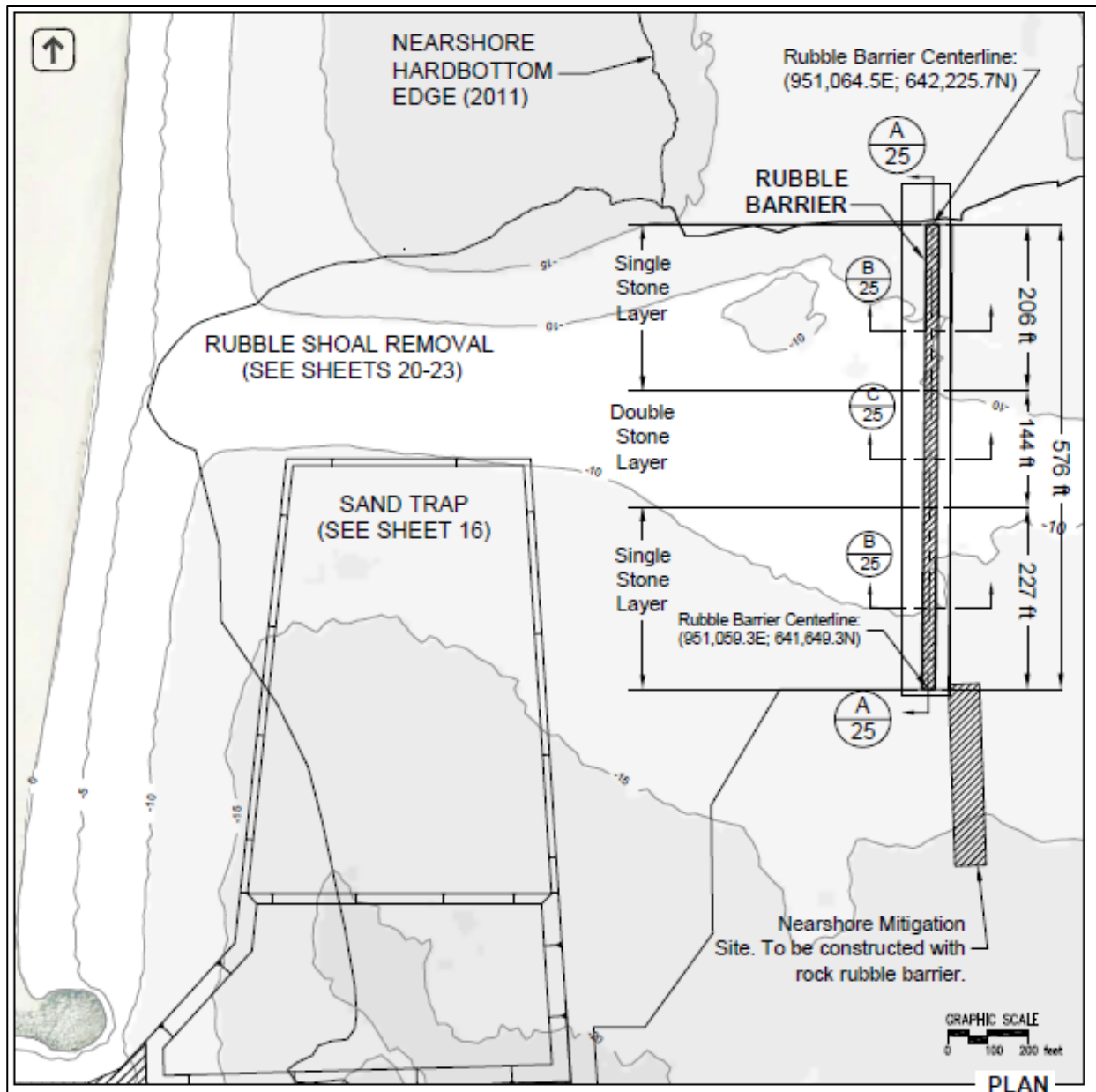


Figure 6. Project area drawing depicting the 577-foot long by 16-foot wide rock rubble barrier to be constructed at the eastern portion of the spoil shoal removal area (situated north to south).

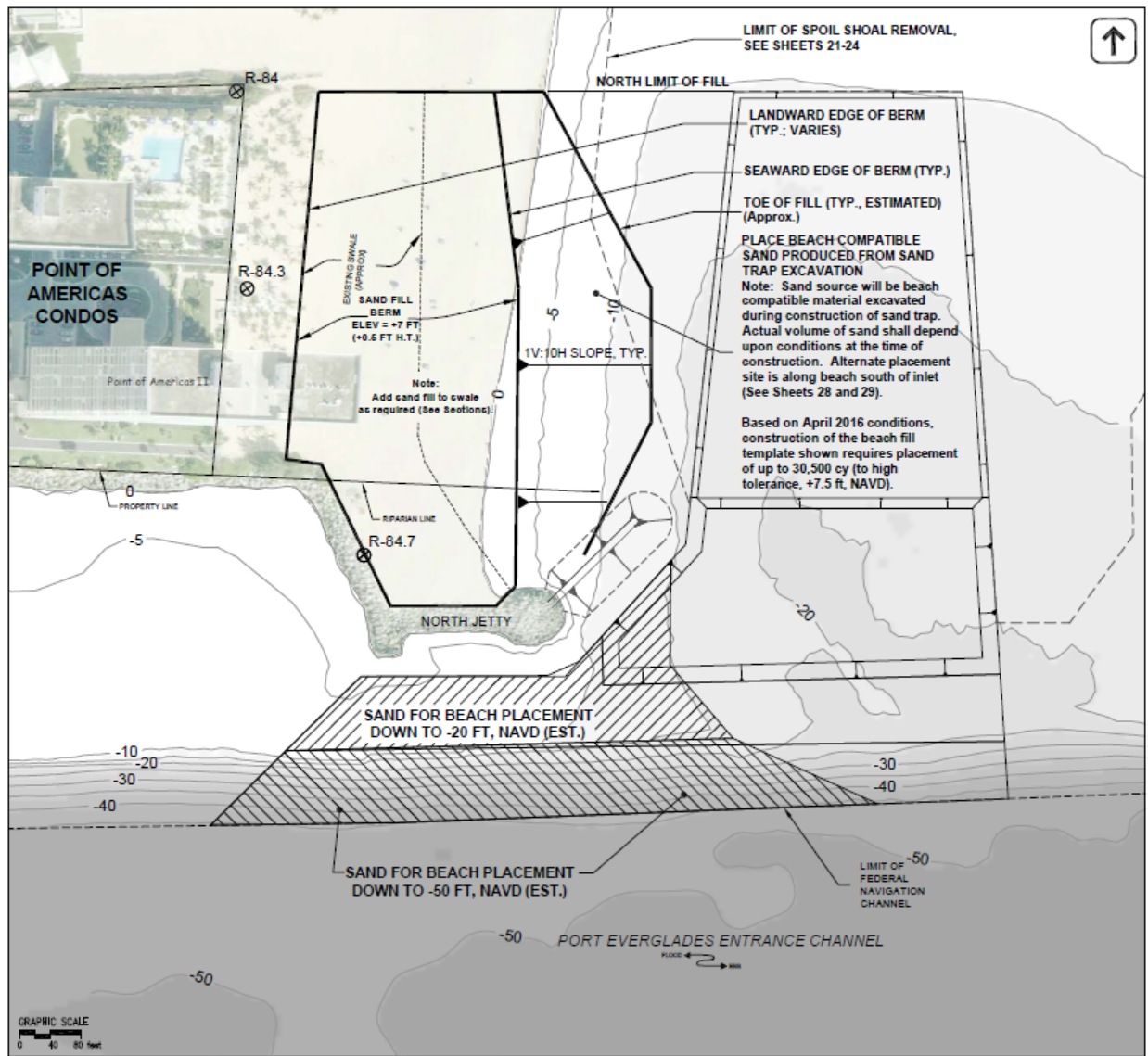


Figure 7. Project area drawing depicting the permitted fill template to the north of the north jetty where beach compatible sediment excavated from the sediment impoundment basin during initial construction (excavation) may be placed.

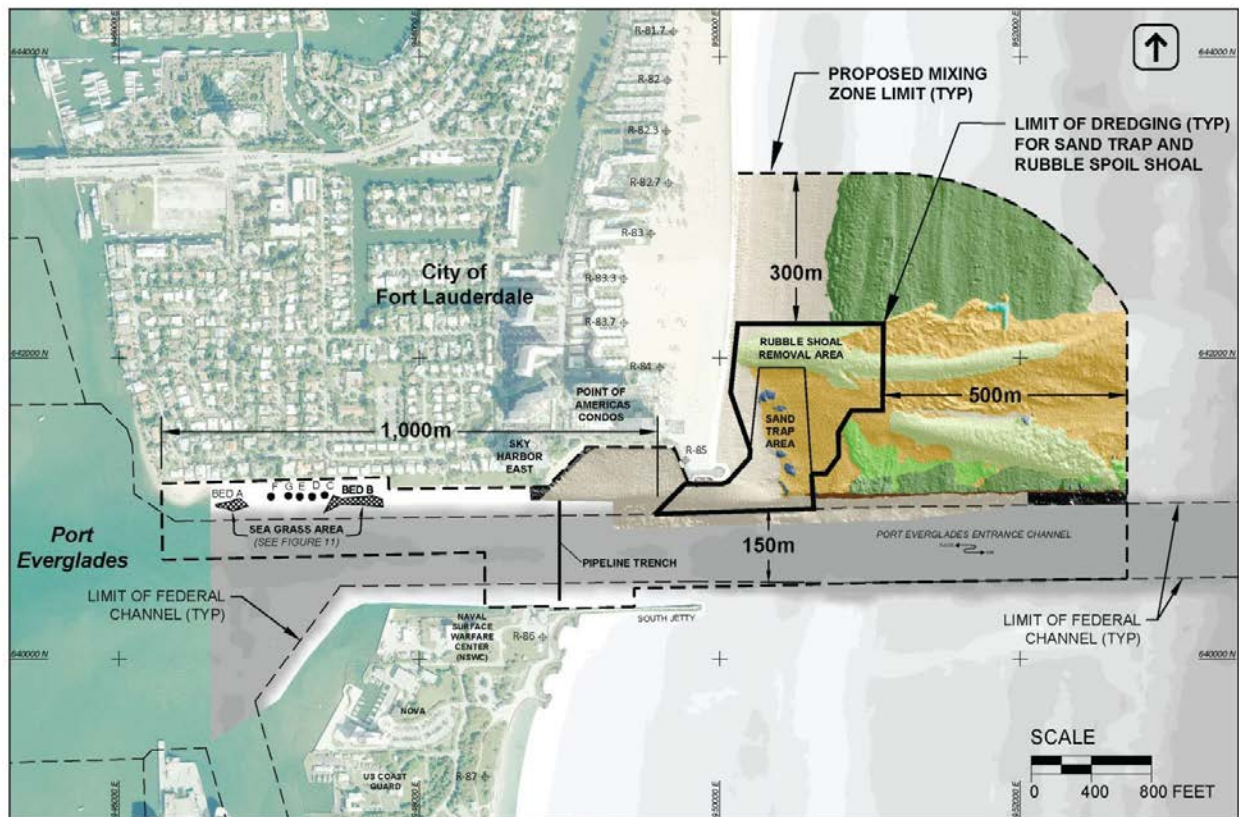


Figure 8. Port Everglades Inlet Channel project area benthic habitat map with direct impact areas (10.8-acre sediment impoundment basin and western portion of the rubble spoil shoal removal area) and indirect impact areas (within the turbidity mixing zone limits of 300 m north, 500 m east, and 1,000 m west) outlined.

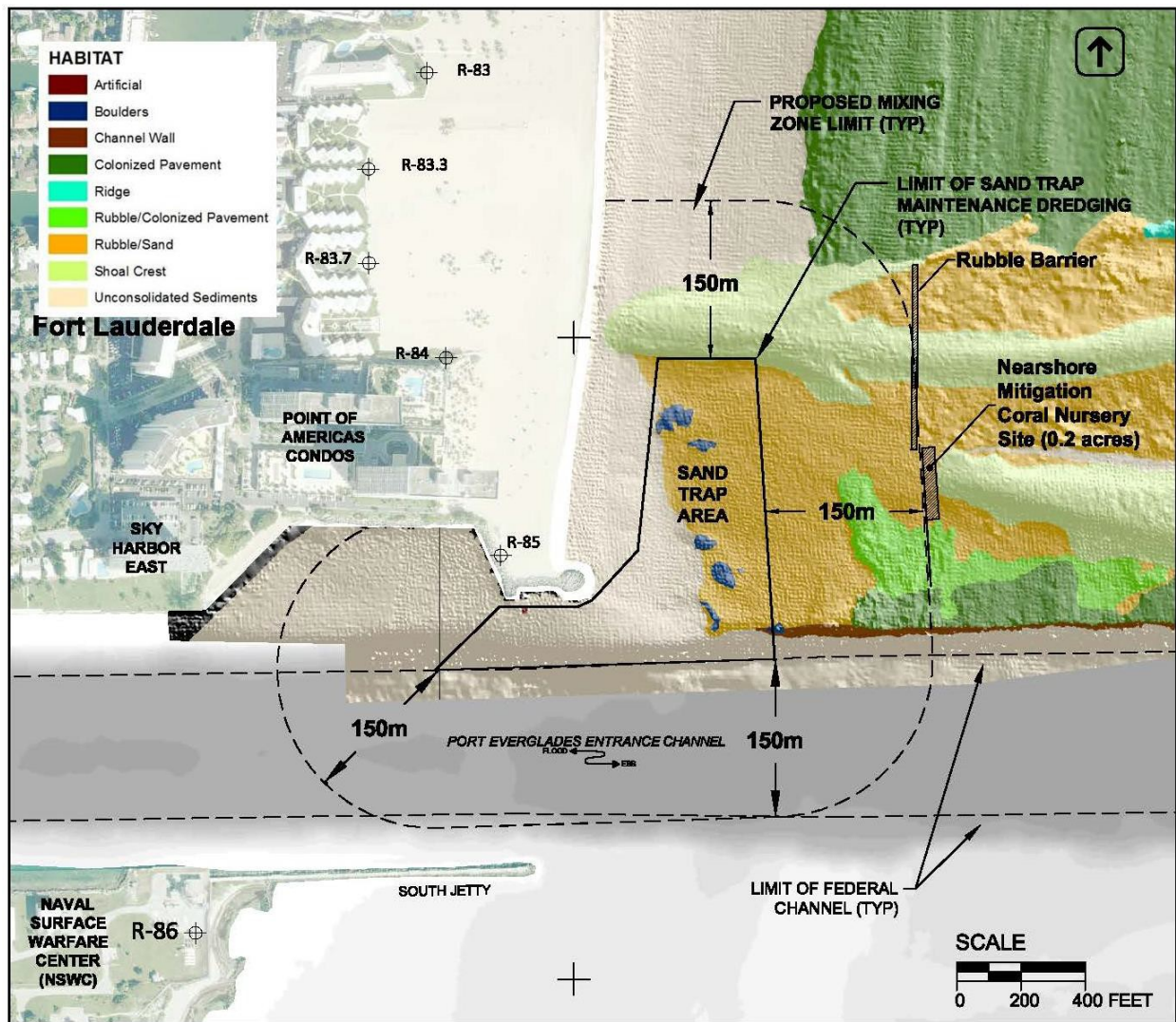


Figure 9. Location of the 0.2-acre nearshore coral nursery site. The 0.2-acre location is shown as the hatched polygon immediately east of the 150-meter mixing zone boundary for maintenance dredging.

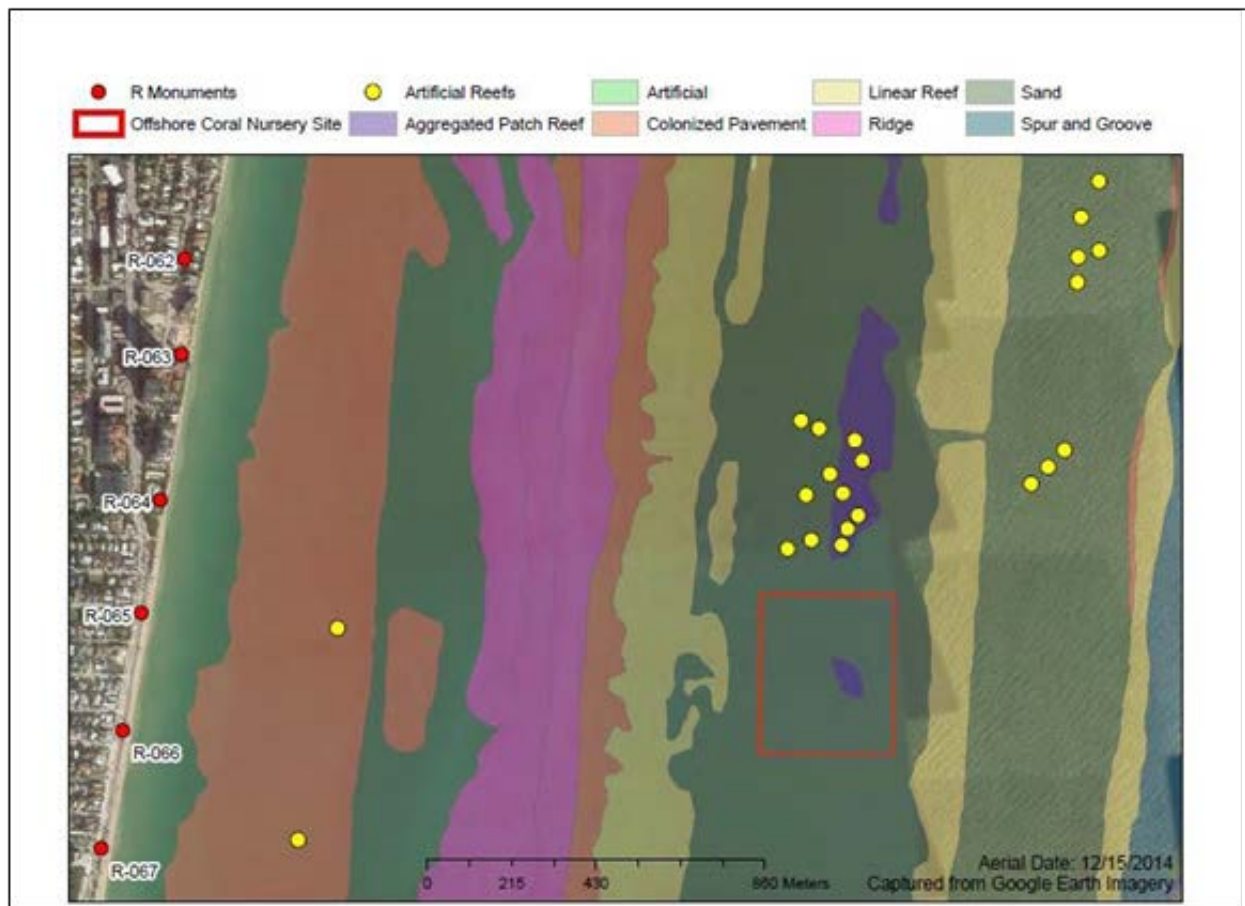


Figure 10. Location of the 0.3-acre offshore coral nursery site in Fort Lauderdale in water depths of approximately 40 ft (12 m).

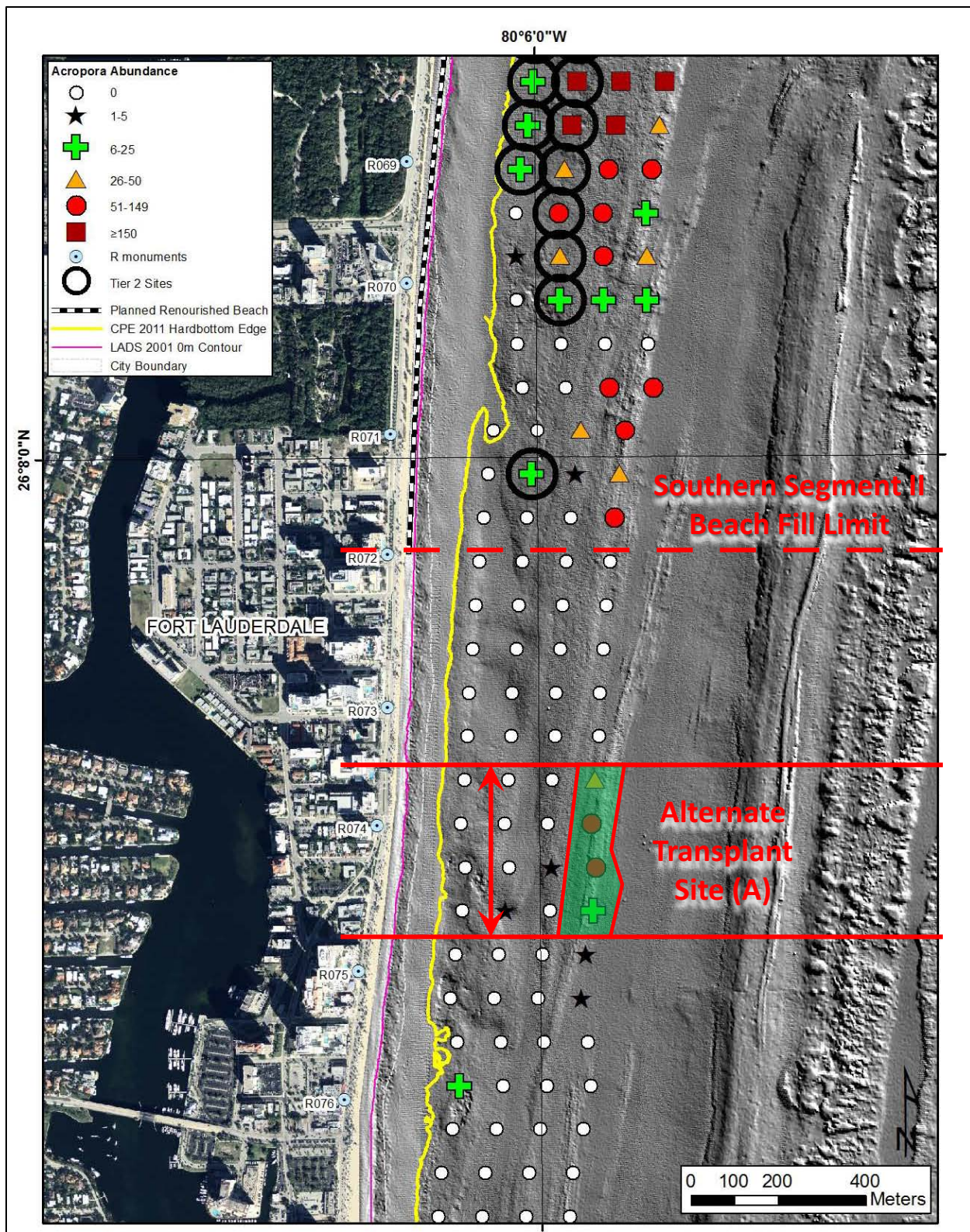


Figure 11a. Approximate location and extent of alternate transplant site (A) for colonies of *Acropora cervicornis* and other federally-listed species of scleractinian corals relocated from the direct and indirect impact areas.

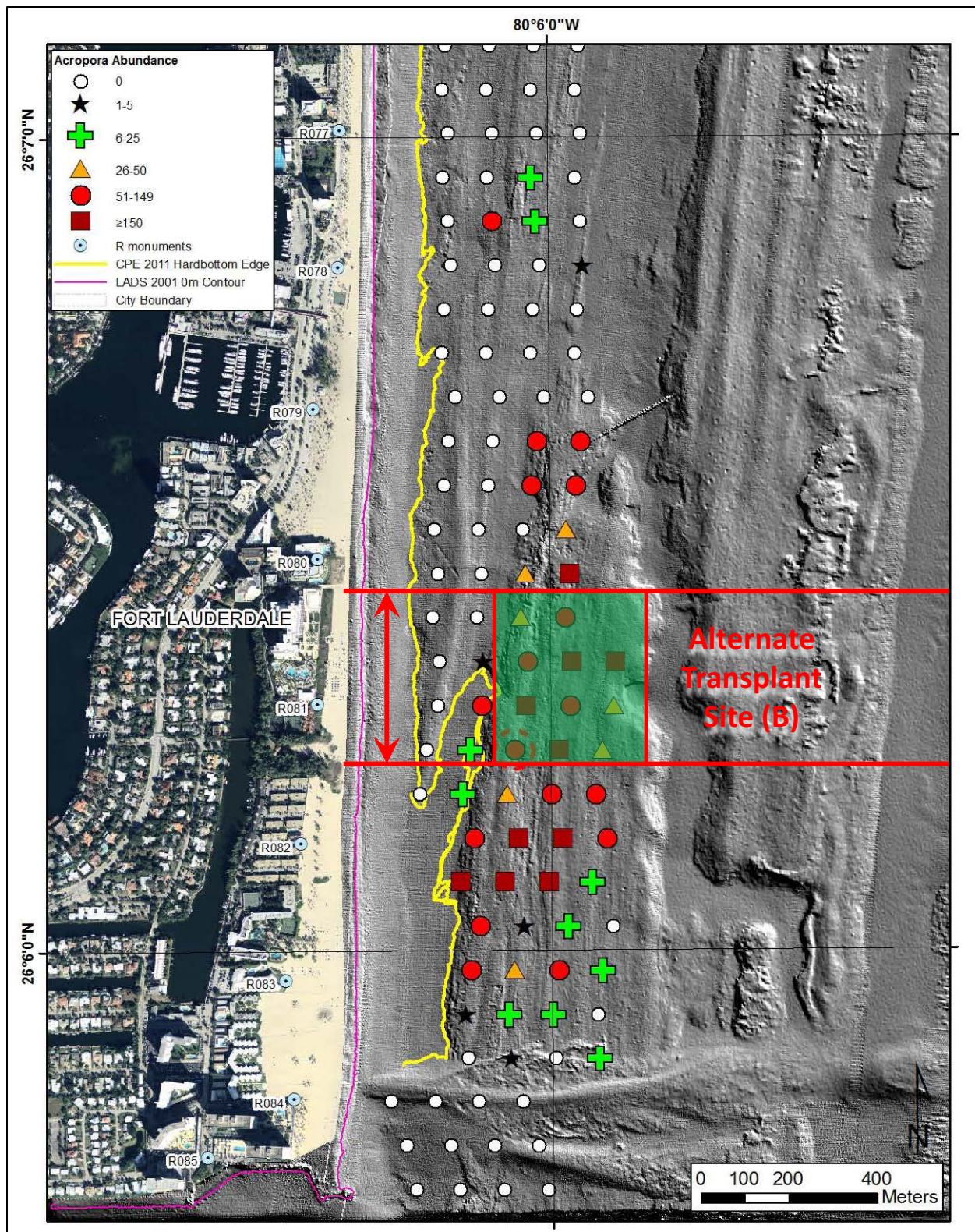


Figure 11b. Approximate location and extent of alternate transplant site (B) for colonies of *Acropora cervicornis* and other federally listed species of scleractinian corals relocated from the direct and indirect impact areas.

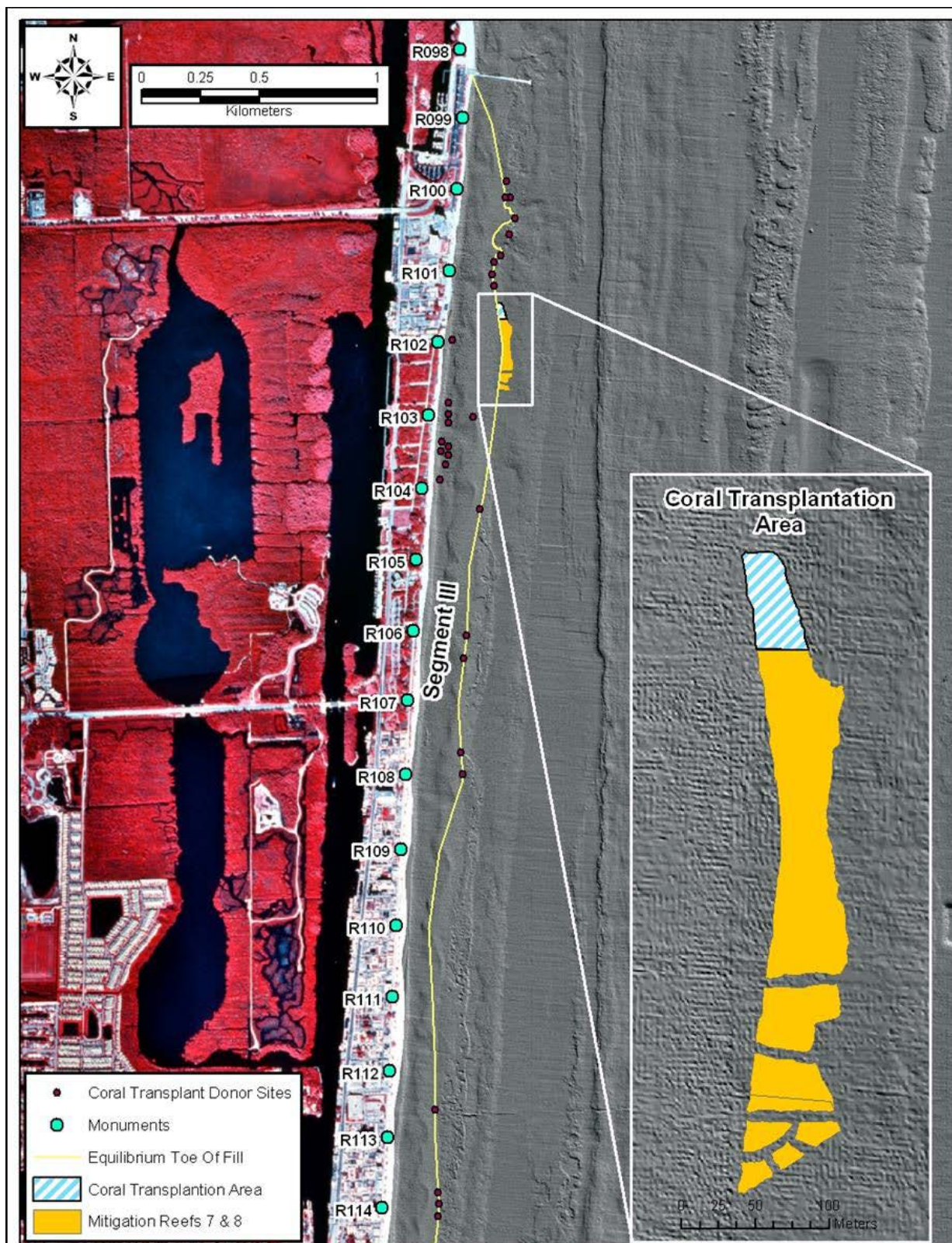


Figure 12. Map showing the location of the transplantation site on the Broward County Segment III mitigation artificial reef (boulder reef) for colonies of octocorals and non-federally listed species of scleractinian corals relocated from the direct impact area.

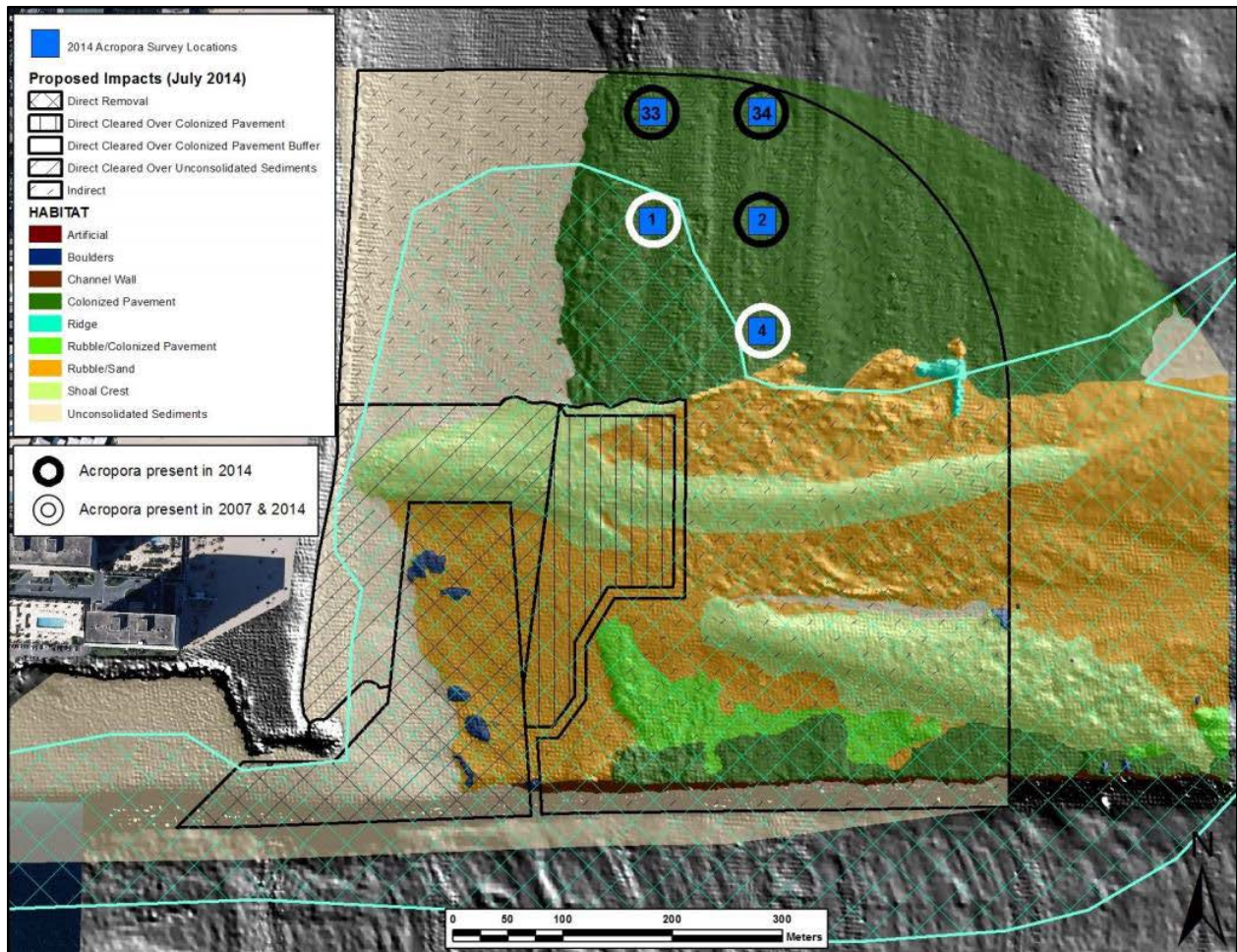


Figure 13. Project area habitat map with the locations of the two sites (1 and 4) from which colonies of *Acropora cervicornis* and any other federally listed species of scleractinian corals will be relocated prior to excavation of the sediment impoundment basin. ESA listed scleractinians at sites 2, 33, and 44 will be monitored under federal requirements (i.e., not dealt with in BMP).

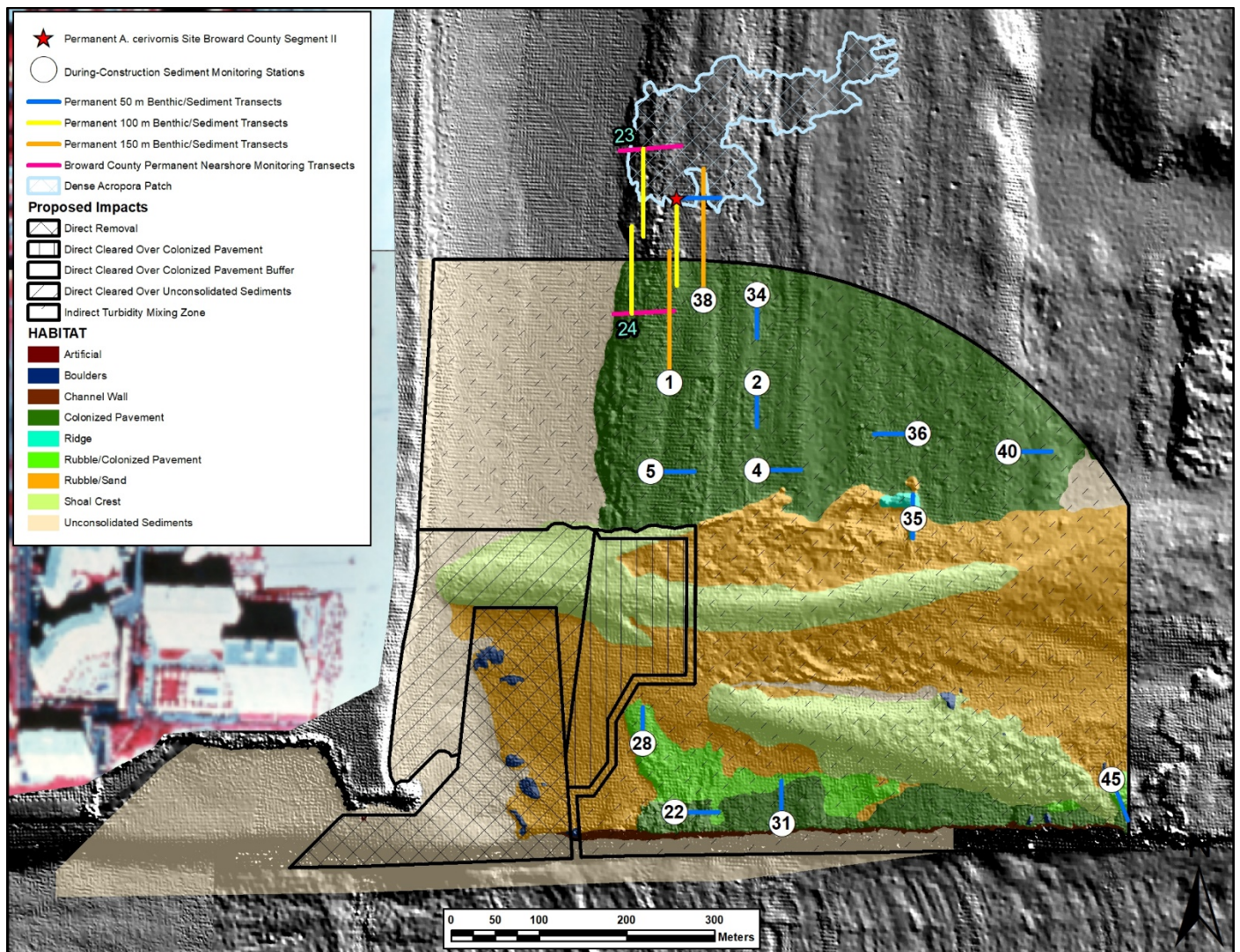


Figure 14. Permanent monitoring stations within and adjacent to the turbidity mixing zone for the Port Everglades Sand Bypass Project. Station numbers in white circles (stations solely within the sand trap turbidity mixing zone) correspond to labels from surveys conducted in 2007 and 2014. The pink, 70 m long, west-east oriented transects at monitoring stations 23 and 24 are also monitored in conjunction with Broward County’s nearshore hardbottom monitoring program. The red star indicates the location of an *Acropora cervicornis* monitoring station that is monitored in conjunction with the Broward County Segment II nourishment project.

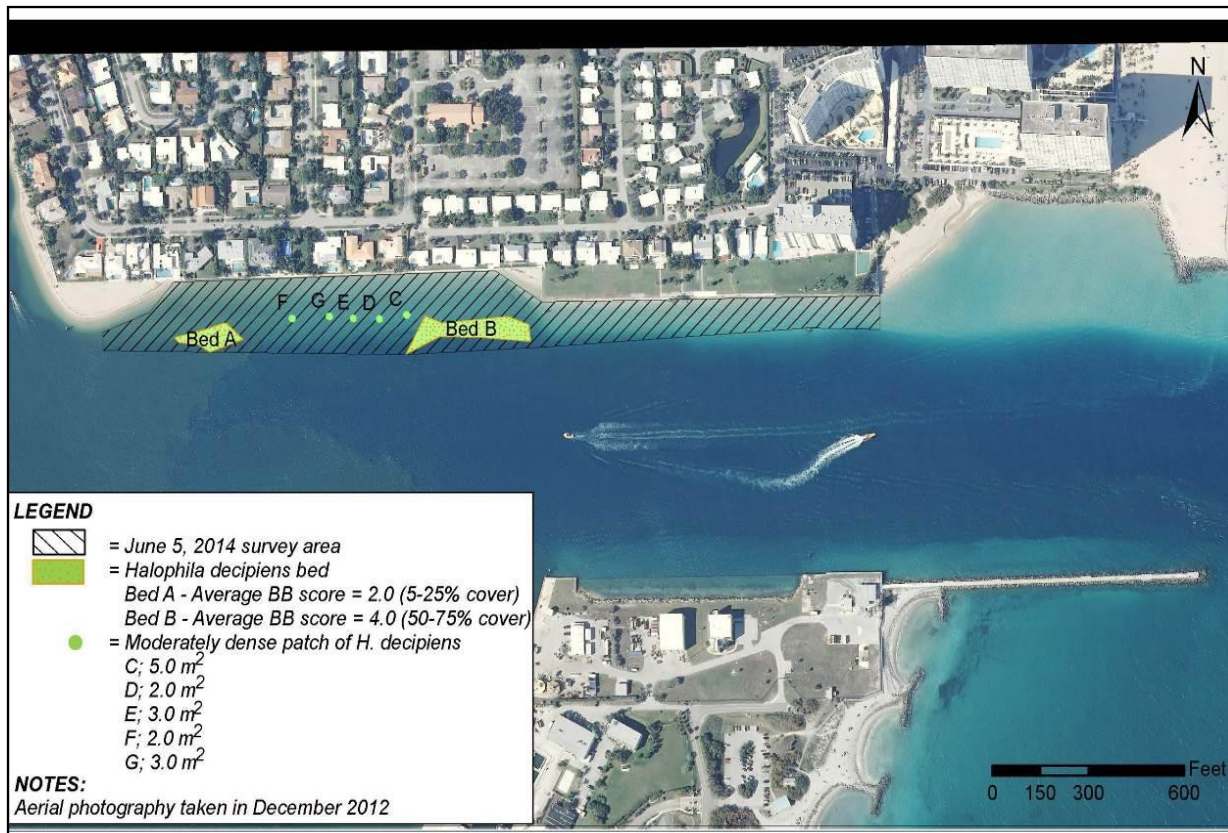


Figure 15. June 2014 seagrass beds on the north side of Port Everglades Inlet Channel within the proposed 1,000-m west turbidity mixing zone.

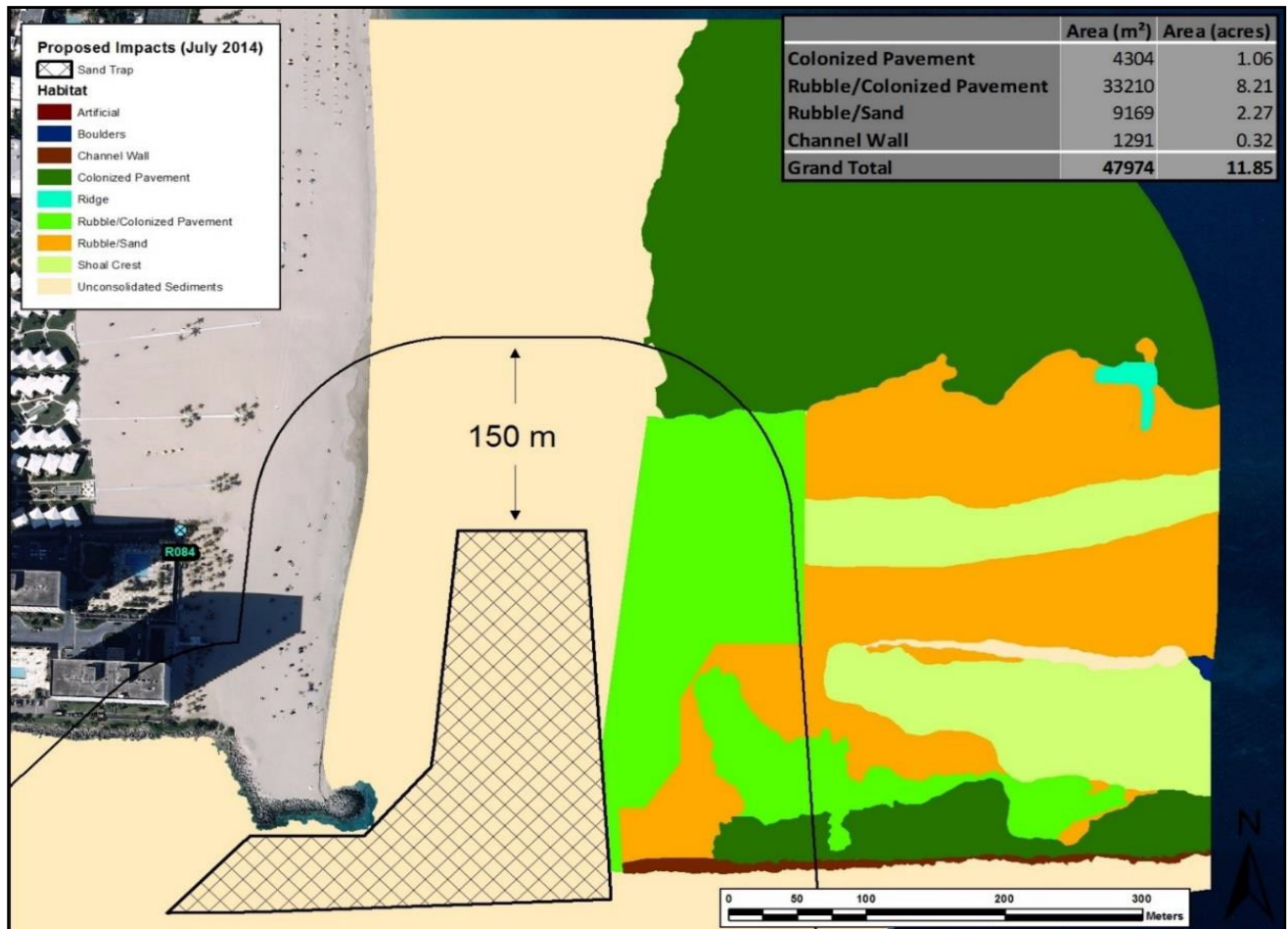


Figure 16. Standard 150-m mixing zone around the sand trap with expected post-project habitats following partial removal of the spoil shoal.

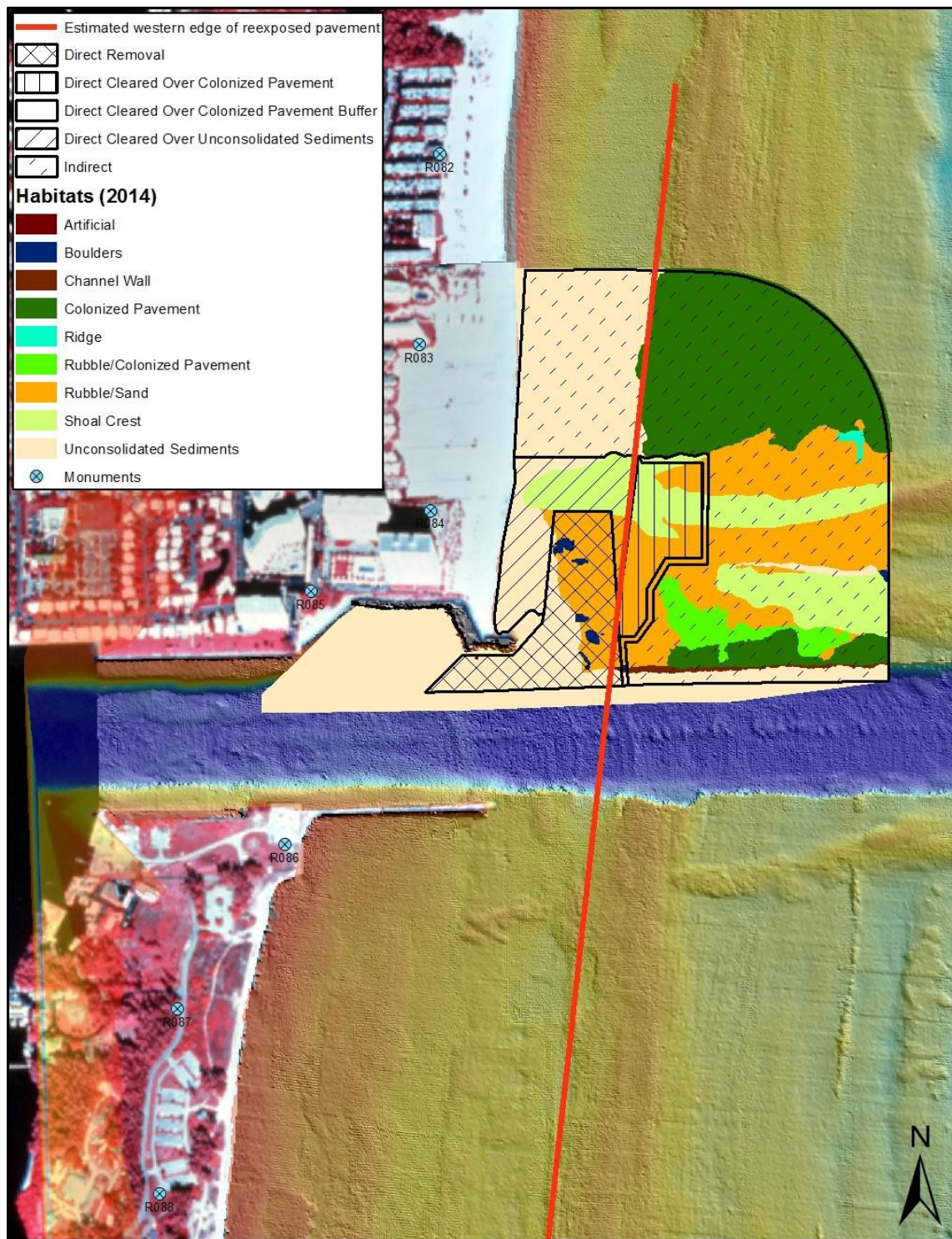


Figure 17. Project area benthic habitat map and adjacent seafloor bathymetry illustrating the line (red) drawn to estimate the western edge of the Colonized Pavement covered by a portion of the Shoal Crest, Rubble/Sand, and Boulder habitats. The line follows the 5.5 m (18ft) depth contour along the same feature from north to south. The benthic habitat limits shown are the 2014 map limits which extend 300 m east of the direct impact area.

- Permanent 150-m Benthic Transects
- Permanent 150-m Sediment Transects
- July 2018 Diver
- Mapped Hardbottom Edge
- 2005 Segment III ETOF
- DNR Monuments

Aerial Date: December 15, 2017

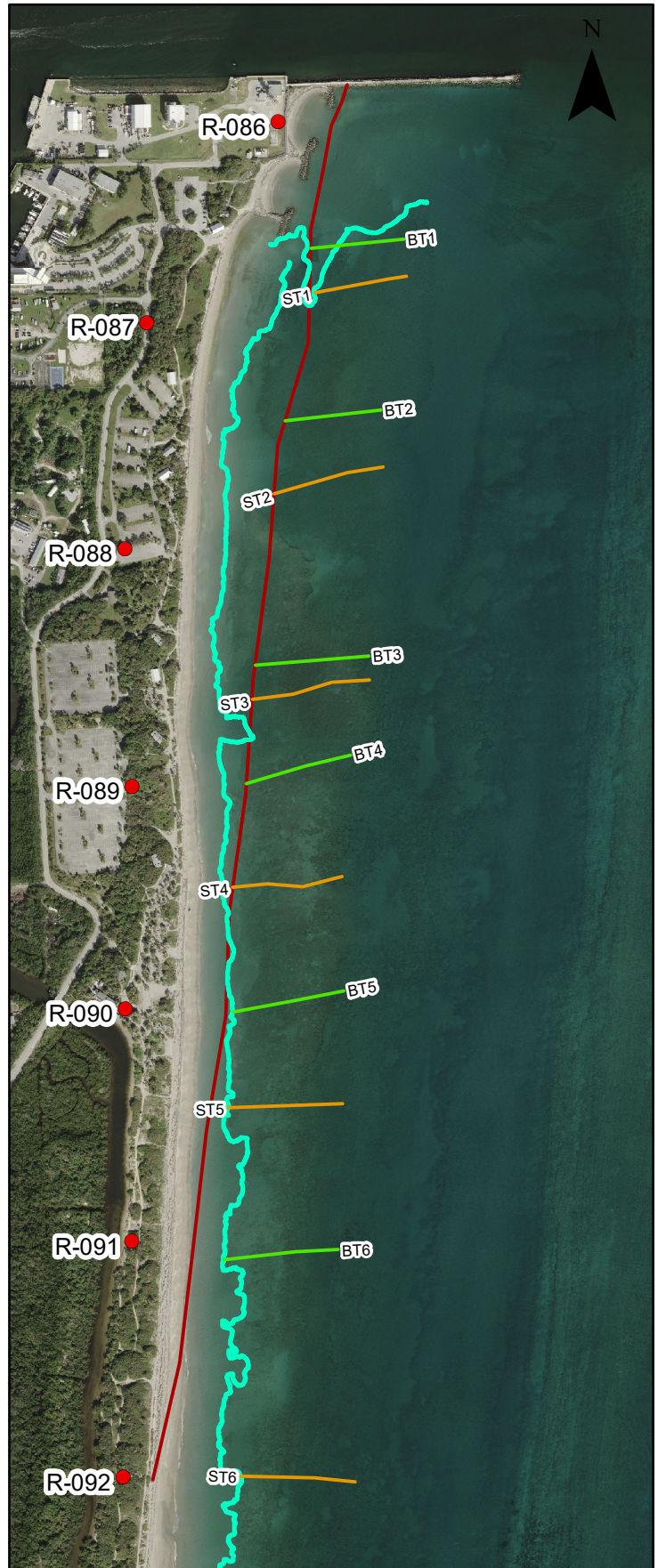
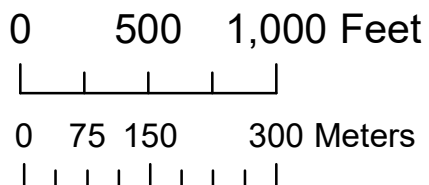


Figure 18. Locations of permanent biological and sediment only monitoring transects adjacent to the fill template (between R-86 and R-92) at the Dr. Von D. Mizell-Eula Johnson State Park.

9.0 LITERATURE CITED

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NSUOC. 2015a. Broward County Port Everglades Sand Bypass Project: Benthic Habitat Mapping and Characterization Update. Draft Final Report. June 2015. Prepared for Broward County Natural Resources Planning and Management Division.

NSUOC. 2015b. Broward County Port Everglades Sand Bypass Project: Benthic Resource Impact Summary. Draft Final Report, June 2015. Prepared for Broward County Natural Resources Planning and Management Division.

Olsen Associates, Inc. (OAI). 2015. Port Everglades Inlet Sand Bypass. FDEP JCP Application File No. 0289308-004-JC

APPENDICES

Appendix A. Impact Avoidance/Minimization Monitoring (Section 2.0) Summary

Project Area	Organisms	*Activities	Monitoring	**Monitoring Deliverables	Plan Sections
Direct Impact Area	<i>Acropora cervicornis</i>	Locate (in 50 m diameter plots), collect, transport and transplant all colonies. Monitor colonies at receiver sites over time.	Follow 25% of colonies (colonies within circular plots). Four (4) monitoring events: at transplantation (t=0) and at 3, 6, and 12 months post-transplantation. At each event record size and colony condition.	Raw Data (PDF's of Field Sheets and Excel Spreadsheets) and Reports.	2.1, 2.1.1.1, 2.1.1.2, 2.1.1.3.1, 2.1.1.4.1. and 7.0
	Non-Acropora Scleractinian Corals and Octocorals	Locate (in 50 m diameter plots), collect, transport, and transplant scleractinians > 10 cm in diameter and Octocorals > 15 cm in height. Monitor coral colonies at receiver sites over time.	Follow 25% of colonies (mapped colonies on tagged boulders). Four (4) monitoring events: at transplantation (t=0) and at 6, 12, and 36 months post-transplantation. At each event record size and colony condition.	Raw Data (PDF's of Field Sheets and Excel Spreadsheets) and Reports.	2.1, 2.1.1.1, 2.1.1.2, 2.1.1.3.2, 2.1.1.4.2, and 7.0
	<i>Strombus gigas</i> (Queen Conch)	Locate (in 50 m diameter plots), collect, and relocate individuals to nearshore hardbottom outside ETOF at Dr. Von D. Mizell-Eula Johnson State Park.	N/A	N/A	2.1, 2.1.2
	<i>Diadema antillarum</i> (long-spined sea urchins)	Locate (in 50 m diameter plots), collect, and relocate to either <i>A. cervicornis</i> receiver sites or non-Acuporid scleractinian coral receiver sites.	N/A	N/A	2.1, 2.1.2
Indirect Impact Area	Federally listed scleractinian corals	Locate (within 2 sites [1 & 4]), collect, transport, and transplant listed scleractinians > 5 cm in diameter. Monitor colonies at receiver sites over time.	Follow 25% of colonies (mapped colonies on tagged boulders). Four (4) monitoring events: at transplantation (t=0) and at 3, 6, and 12 months post-transplantation. At each event record size and colony condition.	Raw Data (PDF's of Field Sheets and Excel Spreadsheets) and Reports.	2.2, 2.2.1 - 2.2.4, and 7.0

* Locating, collecting, transporting, and transplanting activities as well as initial (baseline) monitoring of scleractinians and octocorals shall occur and be completed prior to construction (excavation) of the sediment impoundment basin.

** Reporting requirements for data and monitoring reports are provided in **Section 7.0** of the Plan.

Appendix B. Sand Trap Excavation Event Mixing Zone Monitoring (Section 3.1) Summary

Monitoring Type	Project Area	Survey Unit(s)	Survey	*Monitoring Deliverables	Plan Sections
Sedimentation Monitoring for Hardbottom	Within Turbidity Mixing Zone	N=13 Monitoring Stations. N=1 Transect per Station. - 50 m Transects at Station No's 2, 4, 5, 22, 28, 31, 34, 35, 36, 40, and 45; - 150 m Transects at Station No's 1 & 38.	Interval Sediment Depth Measurements - Pre, During, & Post-construction	Excel spreadsheet, PDF of field sheets	3.1.1.2
			Quantitative Photographs - Pre and Post- construction	Photos, Point Count Files, Excel spreadsheet	3.1.1.3
			Coral and Octocoral Survey within 50 m x 1 m Belt Transect. - Pre and Post- construction	Excel spreadsheet, PDF of field sheets	3.1.1.4
			Qualitative Video - Pre-construction; During and Post-construction if triggered	Video	3.1.1.5
			Important: Sedimentation Impact Triggered Changes to Monitoring		3.1.1.6
	Adjacent to (and slightly within) Turbidity Mixing Zone	N=3 Monitoring Stations (No's 23 and 24 and <i>A. cervicornis</i> station). N=2 Transects per station.	Interval Sediment Depth Measurements - Pre-construction; During and Post-construction if triggered	Excel spreadsheet, PDF of field sheets	3.1.2.2
			Quantitative Photographs - Pre-construction; Post-construction if triggered	Photos, point Count Files, Excel spreadsheet,	3.1.2.3
			Qualitative Video - Pre-construction; During and Post-construction if triggered	Video	3.1.2.4
			Important: Sedimentation Impact Triggered Changes to Monitoring		3.1.2.5
Additional Turbidity Monitoring for Hardbottom	Within Turbidity Mixing Zone	Water Samples	Samples at 2 locations, 2 times per day (densest portion of plume) - During-construction	Turbidity data	3.1.3, and 3.1.3.1
			Important: Turbidity Trigger – Initiates Sedimentation Inspection		3.1.3.2
		Belt Transect in Inspection Area	If triggered , Inspection of organisms and hardbottom within belt transect. If impacts, installation of belt and monitoring (During and Post-construction).	Shapefiles, Excel spreadsheets, pdf's of Field Sheets, Photos and Video	3.1.3.3
Sedimentation Monitoring for Seagrass (if present)	Within Turbidity Mixing Zone	Patch/Bed	Seagrass Edge Delineation. - Pre and Post-construction; During-construction if triggered	Shapefiles	3.1.4.1
			Turbidity Monitoring - - During-construction	Turbidity data	3.1.4.2
			Important: Turbidity Trigger – Initiates Inspection & Delineation		
			Visual Inspection and Delineation for Impacts - During-construction if Triggered	Notes and Shapefiles	3.1.4.3

* In addition to raw data, deliverables include post-construction monitoring reports. Reporting requirements for data and monitoring reports are provided in **Section 7.0** of the Plan.

Appendix C. Trap Maintenance Dredging Mixing Zone Monitoring (Section 3.2) Summary

Monitoring Type	Project Area	Survey Unit(s)	Survey	*Monitoring Deliverables	Plan Sections
Sedimentation Monitoring for Hardbottom	Within Turbidity Mixing Zone	N=6 Monitoring Stations. N=1 Transect per Station. - All Transects 50 m in length	Interval Sediment Depth Measurements - Pre, During, & Post-construction	Excel spreadsheet, PDF of field sheets	3.2.1.2
			Quantitative Photographs - Pre and Post- construction	Photos, Point Count Files, Excel spreadsheet	3.2.1.3
			Coral and Octocoral Survey within 50 m x 1 m Belt Transect. - Pre and Post- construction	Excel spreadsheet, PDF of field sheets	3.2.1.4
			Qualitative Video - Pre-construction; During and Post-construction if triggered	Video	3.2.1.5
			Important: Sedimentation Impact Triggered Changes to Monitoring		3.2.1.6
			Important: Triggers for Terminating and Reinstating Monitoring		3.2.1.7

* In addition to raw data, deliverables include post-construction monitoring reports following each maintenance dredging event. Reporting requirements for data and monitoring reports are provided in **Section 7.0** of the Plan.

Appendix D. Mitigation Monitoring (Section 4.0) Summary

Mitigation Type	Project Area	Survey Unit(s)	Survey / Activity	*Monitoring Deliverables	Plan Sections
Re-exposure of Hardbottom	Re-exposed Colonized Pavement Habitat	Re-exposed Colonized Pavement Area	Post-construction Analysis of Current Aerial Photographs (Map revision). - Post (Years 1, 2, 3, and 5)	Updated Map / Shapefiles, Acreage Estimation	4.1.1.1.1
			In-water Ground-truthing and Boundary Delineation. - Post (Years 1, 2, 3, and 5)		4.1.1.1.2
	Re-exposed and Reference Area Colonized Pavement Habitat	N=16 Sites. N=3 Transects per Site. - N=8 Sites in Re-exposed Area. - N=8 Sites in Reference Area.	Quantitative Photographs (one side of each transect). - Post (Years 3 and 5)	Photos, Point Count Files, Excel spreadsheet	4.1.1.2.2
			Scleractinian Coral and Octocoral Quadrat/Belt Transect Survey (both sides of middle transect). - Post (Years 3 and 5)	Excel spreadsheet, PDF of field sheets	4.1.1.2.3
			Important: Trigger for Terminating Monitoring		4.1.2
			Important: Success Criteria	Data and Reports	4.1.3 & 7.0
Boulder Reef Scleractinian Coral Nursery	North and South Mitigation Reef Sites	Mitigation Reefs (N=2) As-built Survey to Document Net Acreage.	Mitigation Reef Boundary Delineation. - Within 6 months of construction	Shapefiles, Gross Acreage	4.2.2.2
			Line-Intercept for Percent Hardbottom. - Within 6 months of construction	Excel spreadsheet, PDF's Field sheets, Net Acreage	4.2.2.3
			Important: Success Criteria	Data and Report	4.2.3 & 7.0

* In addition to raw data, deliverables include post-construction monitoring reports following each maintenance dredging event. Reporting requirements for data and monitoring reports are provided in **Section 7.0** of the Plan.

Mitigation Monitoring (Section 4.0) Summary Continued

Mitigation Type	Project Area	Survey Unit(s)	Survey / Activity	*Monitoring Deliverables	Plan Sections
Relocation of Corals of Opportunity to Coral Nursery Mitigation Reefs and Active Management of Mitigation Nursery Reefs	Spoil shoal and various other locations. Coral Nursery Mitigation Reefs (N=2).		Locate, collect, transport, and transplant at least 3,000 scleractinians corals of opportunity to mitigation reefs.		4.3.1
		Cohorts of Coral Colonies on Boulders	Monitor Survival and Health of Transplanted Colonies. Follow 25% of colonies in each cohort on each reef. Three (3) monitoring events per cohort: at transplantation (t=0) and at Years 1 and 3 post-transplantation.	Excel spreadsheet, PDF of field sheets, and photos (if taken)	4.3.2.1
			Monitor Reproductive Status of Transplanted Colonies. Follow 10% of colonies in each cohort on each reef. Two (2) monitoring events per cohort: at transplantation (t=0) and at Year 3 post-transplantation.	Excel spreadsheet, PDF of field sheets, results of histological analysis	4.3.2.2
	Coral Nursery Mitigation Reefs (N=2).	Area within 25 cm of all relocated colonies on tagged boulders	Clean Boulders at Mitigation Reefs. Wire brushes used to remove turf and macroalgae from 25 cm diameter areas (circles) around each transplanted coral colony. - Cleaning events: Years 1-6 post construction of each reef.		4.3.3.1
			Track recruitment of scleractinian corals to Mitigation Reefs. Record all recruits > 2 cm in max dimension. - Monitoring events: annually for duration of Survival and Health monitoring of Transplanted colonies.	Excel spreadsheet, PDF of field sheets, and photos (if taken)	4.3.3.2
	Rubble habitats at spoil shoal and Mitigation Reefs (N=2).		If urchins are present and abundant enough, Relocate urchins (<i>Diadema antillarum</i>) from Rubble habitats at spoil shoal to Mitigation Reefs. - Relocation events: Years 1 – 3 post-construction of each reef if possible.		4.3.3.3
			Important: Success Criteria	Data and Report	4.3.4 & 7.0

* In addition to raw data, deliverables include post-construction monitoring reports following each maintenance dredging event. Reporting requirements for data and monitoring reports are provided in **Section 7.0** of the Plan.

Appendix E. Beach Fill Placement Monitoring (Section 5.0) Summary

Project Area	Survey Unit	Survey	Frequency	*Monitoring Deliverables	Plan Sections
Fill Placement Area (R-86 to R-92)	Permanent 150 m Transects (N=12); 6 Biological Transects and 6 Sediment Only Transects	Line-Intercept: along all 12 transects	Pre-construction (N=1): Once prior to initial fill placement event. Post-construction (N=4 per fill placement event): Immediately (within 6 months) and annually for 3 years (years 1, 2, and 3) following each fill placement event.	Excel spreadsheet, PDF of field sheets	5.2.2
		Interval Sediment Depth Measurements: along all 12 transects		Excel spreadsheet, PDF of field sheets	5.2.3
		Quadrat Sampling: within 1 m ² quadrats along 6 Biological transects		Excel spreadsheet, PDF of field sheets	5.2.4
		Qualitative Video: along 6 Biological transects		Video	5.2.5
	Hardbottom Edge	Mapping of Landward Nearshore Hardbottom Edge: between R-86 and R-92		Shapefiles	5.1